CHAPTER 4

Panic Disorder, Social Anxiety Disorder, and Phobias

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Go to the heart of danger . . . for there you will find safety.
—Old Chinese proverb

You must do the things you think you cannot do.
—Eleanor Roosevelt

Four types of anxiety disorders are discussed in this chapter: (1) panic disorder, typified by recurrent attacks that involve the rapid onset of multiple physical symptoms (e.g., shortness of breath and racing heart), as well as fear of future attacks; (2) agoraphobia, characterized by avoidance of situations in which anxiety or panic symptoms might occur, especially when escape may be difficult; (3) social anxiety disorder (previously termed "social phobia"), which involves intense anxiety or avoidance of interpersonal and/or performance situations fueled by fear of negative evaluation by others; and (4) specific phobias, characterized by excessive fear toward a particular object or situation (e.g., animals), certain natural environments (e.g., heights), and closed spaces.

The anxiety disorders overlap considerably with one another, both in terms of their typical symptoms and the models of etiology to explain their development. Thus, rather than review all of the data for each disorder separately, we highlight those aspects of panic disorder and phobias that are shared, then consider the evidence specific to a given disorder. We outline each disorder's common symptoms and related clinical features, natural course and treatment outcome findings, epidemiological data, and the high rates of comorbidity for panic and phobic disorders.

Finally, we review general and specific models of etiology, emphasizing the many shared biological and psychological vulnerability factors across the anxiety difficulties. Throughout the chapter, we raise potential clinical implications that follow from the psychopathology literature and consider some of the many exciting challenges that arise in helping patients reduce their disabling anxiety symptoms.

TYPICAL SYMPTOMS AND ASSOCIATED CLINICAL FEATURES

The anxiety disorders, including panic disorder, social anxiety, and the various phobias, share a number of features, such as avoidance behaviors and excessive fear relative to what others would experience in a given situation. These common features (as we discuss below) have led many researchers to suggest that a dimensional perspective may more appropriately characterize the range of anxiety problems rather than the current classification system that emphasizes specific categories. Relatedly, certain transdiagnostic models (e.g., Nolen-Hoeksema & Watkins, 2011) integrate distal (e.g., childhood sexual abuse) and proximal (e.g., rumination) risk factors, as well as moderators (e.g., social stress), to help explain why a given risk factor may lead to multiple *different* disorders (this concept is known as "multifinality"). Furthermore, these models can help to explain why one individual with a given risk factor may develop one set of symptoms, another individual with the same risk factor may develop a different set of symptoms, and a third individual may not have any significant mental health symptoms (this concept is known as "divergent trajectories"). Thus, we consider symptom dimensions that are common across panic, social anxiety, and phobic disorders, as well as the unique ways these difficulties play out across specific disorders. This focus on both common and disorder-specific features is important from a treatment perspective because seeing patients with multiple rather than isolated anxiety difficulties is the norm (see Brown, Campbell, Lehman, Grisham, & Mancill, 2001).

Subjective distress that is excessive given the objective danger in the environment is present across anxiety difficulties, on a continuum from diffuse worry and future-oriented anxious apprehension (often present in agoraphobia), to more specific targeted fears (seen in specific phobia and social anxiety disorder), to imminent threat and panic (characteristic of panic disorder). Related to the *experience* of emotion (e.g., fear, anxiety, distress), there are also impairments in the *regulation* of emotions (Cisler, Olatunji, Feldner, & Forsyth, 2010). For instance, anxious individuals typically strive to make their world feel "safe." These maladaptive strategies to reduce the sense of danger and vulnerability often result in extensive efforts to avoid or escape from the feared situations or feelings, which adds considerably to the impairment in occupational, academic, and social functioning typical of panic, social anxiety, and phobic disorders. Similarly, anxious individuals often rely too heavily on cues or signals in their environment that help to reduce their distress at entering feared situations (see Rachman, 1994). For instance, a person with agoraphobia may only go out to the mall if their spouse accompanies them; a person with panic disorder may insist on always carrying their medication or water bottle when leaving the house; a person with social anxiety disorder may only attend a party after consuming three beers. Finally, although much previous research has focused on *negative*

emotions and emotion regulation, to better understand the experience of those with pathological anxiety, there is an increasing focus on impairments in *positive* emotions and emotion regulation (e.g., Eisner, Johnson, & Carver, 2009; Kashdan et al., 2013). This is reflected in the individual with social anxiety who does not emotionally benefit from savoring the memory of their date and even fears being evaluated positively by their date, or the person with panic disorder who actively dampens positive emotions when they arise.

Anxious individuals also tend to process information in distorted ways that are thought to increase their beliefs about the world being a dangerous, uncontrollable place and the individual being vulnerable to threat (see MacLeod & Mathews, 2012). Examples include spider-fearful individuals orienting their attention quickly to pictures of spiders (e.g., Mogg & Bradley, 2006); individuals who are afraid of dogs being more likely than nonphobic individuals to use ferocity features in classifying dogs (Landau, 1980); people with panic disorder misinterpreting benign changes in bodily sensations as catastrophic signs of illness (e.g., interpreting a slightly racing heart as indicative of an impending heart attack; Teachman, Smith-Janik, & Saporito, 2007); or people with social anxiety disorder judging themselves more harshly on a social performance (e.g., giving a speech) than do outside observers (e.g., Norton & Hope, 2001). Furthermore, there is compelling evidence that threat-oriented cognitions occur both at relatively *explicit* (controlled, deliberative) and *automatic* (outside one's conscious control) levels (Teachman, Clerkin, Cunningham, Dreyer-Oren, & Werntz, 2019). Additionally, there is some indication that individuals with phobic symptoms may actually *see* their environment as more threatening at a perceptual level, in addition to judging it as more dangerous (e.g., Teachman, Stefanucci, Clerkin, Cody, & Proffitt, 2008).

Just as thinking patterns often become highly geared toward managing perceived threats when individuals become anxious, so too can physiological responding. There is considerable variability in the physiological changes that occur across the anxiety disorders, but some degree of the "fight-or-flight" reaction (initially described by Cannon, 1929) is common to both panic and phobic reactions. The fight-or-flight system involves dramatic surges of autonomic arousal that motivate individuals either to escape or avoid a situation or defend themselves. A freezing response may also occur in some individuals. During the heightened activation of the sympathetic branch of the autonomic nervous system, a series of physical reactions that occur very rapidly help to prepare the body for action. These include increased heart rate, rapid breathing, inhibition of digestive processes, constriction of blood vessels in some parts of the body with dilation of blood vessels for muscles, pupil dilation, endorphin release, and acceleration of reflexes (see Gray & McNaughton, 1996; Gunnar & Quevedo, 2007). This response prompts the individual for immediate action and is adaptive when a real danger exists in the environment, but it essentially (mal) functions as a false alarm when no objective danger exists, as occurs in panic and phobias (see Barlow, 2002).

Thus, across a variety of emotional response systems (e.g., subjective distress, behavior, physiology, cognitive processing; Eysenck, 2014; Lang, 1985), we see common signs and consequences of anxious arousal associated with both panic disorder and phobias, pointing to their shared phenomenological presentation. These common features (combined with frequent difficulties in determining the clinical

threshold for a given disorder, as well as the high level of comorbidity across them) have led to many questions about the adequacy of a categorical classification system for these disorders (see review in Barlow, 2002, Chapter 9). Simply put, questions have arisen as to whether the current system effectively "carves nature at its joints." However, the heuristic value of categories (e.g., for communication about diagnoses) and advantages conferred by the specificity of categorical criteria for research and clinical decision making make it unlikely that the categories will be fully abandoned any time soon (see Brown & Barlow, 2009).

Taking into consideration the overlap across anxiety disorders also recognizes that the typical symptoms unique to each disorder represent important variation in the way anxiety pathology manifests across individuals, depending on their particular vulnerability factors and learning experiences. We turn to these specific symptoms next.

Panic Disorder and Agoraphobia

Panic disorder is defined by recurrent and unexpected panic attacks, which are followed by at least 1 month of fear about future panic attacks, concern about the consequences of the attacks (e.g., worries about "going crazy"), or changes in behavior related to the attacks (e.g., avoidance or introduction of extensive safety cues). The term "panic attack" is used frequently by the general public, but it has a specific set of criteria in the context of anxiety pathology: The attack must involve a distinct period of intense discomfort and/or fear that rises to a peak within minutes. It must involve at least four of the following symptoms: "palpitations, pounding heart, or accelerated heart rate; sweating; trembling or shaking; sensations of shortness of breath or smothering; feeling of choking; chest pain or discomfort; nausea or abdominal distress; feeling dizzy, unsteady, light-headed, or faint; chills or heat sensations; paresthesias (numbness or tingling sensations); derealization (feelings of unreality) or depersonalization (being detached from oneself); fear of losing control or going crazy; fear of dying" (list from American Psychiatric Association, 2013, p. 208). Attacks that involve fewer than four symptoms are termed "limited symptom" attacks.

Importantly, panic attacks can occur within the context of many anxiety problems. As one example, a person with social anxiety disorder may experience a panic attack before having to give a speech; however, in these cases the fear focuses on the negative evaluation by others and expected social embarrassment. In panic disorder, the attacks themselves become a problem, as indicated by the fear of future attacks and altered behavior in response to the attacks. For this reason, panic disorder is often described as a "fear of fear" (Goldstein & Chambless, 1978). It is notable that frequent panic attacks are not actually required for the diagnosis, provided that impairment or fear of future attacks is evident. Furthermore, the occurrence of panic attacks alone is not sufficient to warrant a diagnosis of panic disorder. In fact, roughly 11% of adults in the United States experience a panic attack during a given 12-month period, which is clearly much higher than the prevalence rate of actual panic disorder (e.g., American Psychiatric Association, 2013). Thus, tracking and reducing how often panic attacks occur are useful in treatment but not the sole goal of treatment; rather, helping people reduce their fear of future attacks and diminish associated avoidance

is paramount. Along these lines, simply asking patients to monitor the percentage of time they are worried about becoming anxious can provide the therapist—and the patient—with a useful index of change. This marker is meaningful because, at the outset of treatment, the lives of these patients are often very much about their anxiety; indeed, in most of their waking hours, they are preoccupied with concern about becoming anxious.

Panic attacks may be entirely unexpected, in that they seem to come on "out of the blue" (when a person cannot identify a trigger), especially during early stages of the disorder. It has been our clinical experience that more often than not, such uncued attacks occur in situations in which the person is unaware of the precipitating factors (e.g., a fleeting thought and emotion). Alternatively, they may be expected, in that an attack is more likely to occur following certain triggers (e.g., when in a crowded place), or when the attack regularly occurs after exposure to a particular situation. Situational events may involve a place, such as a restaurant, but they may also be a physical state or symptom (e.g., when experiencing arousal following caffeine intake), or an emotional circumstance (e.g., during conflict with significant others). Not surprisingly, people often start to avoid situations or feelings that tend to trigger their panic attacks. When this becomes extreme and impairing in its own right, the person may qualify for diagnoses of both panic disorder and agoraphobia.

Currently in the field, we consider agoraphobia as an independent disorder, even though it often co-occurs with panic disorder. A person with agoraphobia experiences unusually high anxiety about situations or places that they perceive would be hard to escape from, or in which they would not be able to find help if they have a panic attack or similar symptoms. We think of such experiences as a "disorder" when they last for a while (months), and cause a person enough stress that they find themself going out of their way to avoid those situations and places. Common situations that trigger agoraphobic avoidance include various modes of transportation (e.g., car, bus), going outside of the home alone, or standing in crowds or lines. Avoidance of internal sensations (known as "interoceptive cues") that make a person fearful is also common. These include sensations associated with dizziness, shortness of breath, or hyperventilation (Schmidt & Trakowski, 2004), as well as sensations associated with normal arousal in response to exercise, sex, eating heavy meals, or strong emotional reactions.

Given the considerable avoidance behaviors and fear of bringing on panic symptoms that characterize panic disorder and/or agoraphobia, it is not surprising that significant impairment is associated with these disorders. The morbidity associated with panic disorder and agoraphobia cuts across many domains of biopsychosocial functioning. For instance, in terms of relationship functioning, couples may have stopped having sexual intercourse because of the patient's fears about experiencing intense physical sensations. In addition, difficulties with driving and related travel fears often lead to considerable dependence on loved ones. Work impairment is also common (Alonso et al., 2004) as the triggers associated with panic become more ubiquitous. In severe cases of agoraphobia, the person may literally be homebound, which clearly limits work opportunities. Individuals with panic disorder and/or agoraphobia are also at an elevated risk for experiencing a range of other anxiety, mood, impulse-control, and substance use disorders (Kessler et al., 2006). Additionally, sleep dysregulation is common and may be tied in part to the frequent occurrence of

nocturnal panic attacks (Craske et al., 2002). Patients who experience nocturnal panic attacks, which occur when a person wakes from sleep in a state of panic, show similar panic symptom severity and general impairment to that of panic patients who do not experience nocturnal panic. However, patients with nocturnal panic attacks tend to show lower levels of agoraphobic avoidance, possibly because they have more fearful associations with sleep than with external situational factors.

One of the most serious clinical features connected to panic disorder is the increased risk for suicide attempts (see Schmidt, Woolaway-Bickel, & Bates, 2001). In an early meta-analysis (Harris & Barraclough, 1997), panic disorder was associated with a 10-fold increase in suicide risk, relative to expected rates in the population. Moreover, although there is some support for the "depression mediation hypothesis" (i.e., that panic disorder is *indirectly* linked to suicide via its relationship with heightened depression; Schmidt et al., 2001), there is little doubt that clinicians need to be alert to the increased risk in this population. This is especially true given that comorbidity is the norm rather than the exception (Kessler et al., 2006). It is possible that elevated rates of suicide risk among individuals with panic disorder may be tied to beliefs that anxiety sensations are dangerous, potentially indicating a cognitive disaster such as going crazy or losing control of one's mind. Referred to as "anxiety sensitivity cognitive concerns," researchers theorize that these beliefs may amplify the distress brought on by the sensations associated with mental illness, such as difficulty sleeping or concentrating (Capron, Norr, Macatee, & Schmidt, 2013). In turn, after the "amplified" distress reaches a potently high level, suicidal ideation and attempts may occur.

Social Anxiety Disorder

Social anxiety disorder is characterized by marked and persistent fear of social interactions or performance situations in which scrutiny by others is possible (American Psychiatric Association, 2013). Individuals with social anxiety disorder consistently experience anxiety when confronted with social and/or performance situations, and this disproportionate anxiety typically lasts for at least 6 months to qualify for a diagnosis. Given the common tendency by socially anxious individuals to overestimate threat in social situations, individuals suffering from social anxiety disorder need not recognize that their fears are unreasonable or excessive. Typically, individuals with social anxiety disorder avoid feared social or performance situations. Less frequently, social situations are not avoided but are endured with considerable dread. Significant anticipatory anxiety is also common.

Although early theorists believed that social anxiety is caused by deficits in social skills (see Trower, Bryant, & Argyle, 1978), empirical findings are inconsistent regarding the extent to which individuals with social anxiety disorder actually have social skills deficits (e.g., Stopa & Clark, 1993). What is clear is that compared with healthy control participants, people with social anxiety disorder are typically more critical of their social performances, even after accounting for actual differences in social abilities (e.g., Rapee & Lim, 1992).

Recent research has focused on understanding moderators and mediators linking social anxiety with biased self-evaluations. For instance, there is evidence that among socially anxious individuals, a tendency to engage in postevent processing

(i.e., to ruminatively dwell on previous social interactions) is associated with more negative judgments about one's social performance (Brozovich & Heimberg, 2011). Other research points to the importance of experiential avoidance (i.e., attempts to control or avoid one's internal emotions and thoughts; Kashdan et al., 2014) for understanding why social anxiety symptoms may lead to less positive self-evaluations across time (Sarfan, Cody, & Clerkin, 2019). Moreover, research from Wallace and Alden (1997) indicates that individuals with social anxiety disorder often believe that they will be unable to make desired impressions on others, or that they will fail to live up to others' expectations. This fear is tied to one of the defining symptoms at the heart of cognitive models of social anxiety disorder—extreme fears of negative evaluation (e.g., Clark & Wells, 1995). Interestingly, there is also evidence that individuals with social anxiety disorder fear *positive* evaluation (Weeks, Heimberg, & Rodebaugh, 2008).

Many theorists contend that the distinction between social anxiety disorder and the more commonly experienced forms of shyness is a matter of degree rather than kind, with symptoms of social anxiety ranging along a continuum from mild and transient social unease to shyness to social anxiety disorder (see Heimberg & Becker, 2002). Thus, while social anxiety disorder and shyness have many similarities, symptoms of social anxiety disorder are more extreme across a variety of domains, and the vast majority of people (~88%) who report that they are shy would not meet diagnostic criteria for social anxiety disorder (American Psychiatric Association, 2013).

The issue of diagnosis becomes further complicated when one considers the relationship between social anxiety disorder and avoidant personality disorder. Given the striking similarities and high rates of comorbidity between these two disorders (as they were defined in DSM-IV, which mainly guided this research), some theorists believe that there is not a "distinct symptomatic delineation" between them (see Reich, 2001, p. 157), and avoidant personality disorder is sometimes conceptualized as an extreme form of social anxiety disorder. Interestingly, there also does not appear to be strong evidence for the notion that there are meaningful subtypes of social anxiety disorder with distinct symptom profiles (for reviews, see Heimberg, Holt, Schneier, Spitzer, & Liebowitz, 1993; Rapee, 1995).

The subjective anxiety experienced by individuals with social anxiety disorder is often extremely impairing and distressing, as are the physical sensations typically tied to social anxiety, such as heart racing, sweating, gastrointestinal discomfort, blushing, and so forth. Avoidance is also particularly problematic. For instance, a student who fears appearing inarticulate or stupid may avoid raising a question during class; a teenager who fears being rejected may never risk asking someone on a date; a middle-aged man who fears that others will notice his hands shaking or voice trembling may avoid giving presentations at work. In some cases, this avoidance behavior can even lead to employment or financial difficulties (e.g., when individuals pass up promotions at work because the new position would involve increased social interactions). Moreover, experiential avoidance appears to be critically important in differentiating between adults with social anxiety disorder and healthy control participants (Kashdan et al., 2013).

Alcohol problems within the context of social anxiety are also quite high. Epidemiological research indicates that social anxiety disorder and alcohol use disorder commonly co-occur (Grant et al., 2004; Schneier et al., 2010), possibly because some

individuals with social anxiety disorder drink to minimize anxiety when interacting socially. Indeed, there is evidence that coping motives (i.e., drinking to reduce negative affect) mediate the relationship between social anxiety symptoms and drinking problems, particularly for emerging adults (Clerkin, Werntz, Magee, Lindgren, & Teachman, 2014). The presence of social anxiety disorder may also increase the probability of suicide ideation and attempts (Cougle, Keough, Riccardi, & Sachs-Ericsson, 2009), and individuals with social anxiety disorder have lower well-being ratings compared to individuals without a mental illness (Wersebe et al., 2018).

Specific Phobia

Specific phobias are defined by disproportionate and persistent fear toward a particular object or situation that typically lasts at least 6 months to qualify for a diagnosis (see American Psychiatric Association, 2013). Unlike less intense, subclinical fears, a specific phobia must almost always evoke an extreme and immediate fear reaction. Furthermore, the phobic target must be avoided or endured with severe distress and result in some impairment in functioning. Similar to social anxiety disorder, individuals with a specific phobia need not recognize their fear is unreasonable or excessive because of the tendency by many phobic individuals to have exaggerated perceptions of the danger in phobic situations. Note that we focus in this chapter on panic and phobias among adults, but given the relatively high prevalence of phobias in children, the interested reader is encouraged to see excellent summaries of fear and phobias in children and adolescents (e.g., Ollendick, White, & White, 2018).

The overlap in criteria and feared situations for panic disorder, agoraphobia, social anxiety disorder, and specific phobias can make their differential diagnosis a challenge. More generally, determining whether a fear target constitutes a specific phobia or another anxiety problem is sometimes difficult. For example, a fear of dirt can reflect either obsessive–compulsive disorder for a person with contamination obsessions or be part of a specific phobia. Similarly, fear of a particular situation may be tied to a traumatic event in the context of posttraumatic stress disorder or reflect a more circumscribed specific phobia. Distinguishing among these possibilities involves a functional analysis that includes a careful assessment of the triggers and feared consequences associated with encountering the feared object or situation.

In theory, a phobia can be directed toward almost any target, and identifying some of the more esoteric phobias, often by their Greek or Latin names, can be an entertaining party game (e.g., "apeirophobia," a fear of infinity). However, in clinical practice, a smaller set of phobias tends to be represented, and these have been organized into subtypes in recent DSM versions (i.e., animal, natural environment, blood–injection–injury, situational, and other). Although these types have some heuristic value for research and communication purposes, the utility of using subtype specifiers as part of the diagnostic label has been questioned (see Antony & Barlow, 2002).

As with panic disorder and the other phobias reviewed in this chapter, specific phobias are associated with a strong physiological response, characterized by increased heart rate and skin conductance. In fact, a panic attack is a common reaction for persons with specific phobias upon encountering their feared object/situation (Antony, 1994). Although increased arousal is common across various types

of specific phobias, blood–injection–injury phobia has a unique, additional component to its physiological profile that includes a tendency to faint. Another interesting characteristic of certain specific phobias, such as spider, snake, and blood–injury–injection fears, is that patients often feel a strong disgust reaction to their feared object (see review in Cisler, Olatunji, & Lohr, 2009). One theory (which requires more empirical validation), known as the "disease avoidance model," is that disgust serves an adaptive function, in that it promotes avoidance, which helps to prevent the transmission of disease (Davey, 1992). Supporting this theory, meta-analytic work suggests that although disgust proneness is elevated for a range of anxiety disorders, the effect is especially pronounced for individuals whose anxiety symptoms are linked to contagion fears (Olatunji, Armstrong, & Elwood, 2017). Thus, applying transdiagnostic frameworks may be helpful to understand conditions under which disgust proneness leads to one set of symptoms (e.g., spider phobia) versus another (e.g., health anxiety). For instance, someone with a tendency toward disgust proneness may be more likely to develop spider phobia if their environment emphasizes concerns tied to small animals and disease. Furthermore, the avoidance of disgust elicitors to prevent disease may function similarly to the avoidance in panic disorder of bodily sensations to prevent physical illness; both cases involve an oversensitive "false alarm" system that exaggerates the presence of danger.

Finally, for individuals with specific phobias, avoidance is problematic because it can result in limited functioning in social, recreational, and work situations (see Rowa, McCabe, & Antony, 2006), and persons with blood–injection–injury fears often do not adhere to recommended medical regimens (Mollema, Snoek, Ader, Heine, & van der Ploeg, 2001). However, the impairment associated with specific phobias has more often been described as moderate rather than severe (e.g., Becker et al., 2007), in part because it is often possible to avoid encountering one's feared situation/object effectively because of the circumscribed nature of the fear. For instance, organizing one's life to avoid seeing a snake is far easier than trying to avoid all social performance situations or triggers of bodily sensations (as might be desired in social anxiety or panic disorder, respectively). Thus, it is not unusual for people to meet all of the diagnostic criteria for a specific phobia except for the impairment in functioning requirement (e.g., Romano, Tremblay, Vitaro, Zoccolillo, & Pagani, 2001).

Clinical Implications

The typical symptoms associated with anxiety disorders can be manifested in one or more responses systems: behavioral, physiological, expressive emotion, and/or cognitive. We deal with some of the clinical implications associated with each of these domains, considering the behavioral (e.g., avoidance of therapy), physiological (e.g., difficulty tolerating physiological reactions), expressive emotion (e.g., dysregulation of multiple emotions), and cognitive (e.g., beliefs about not feeling in control, selective processing of threatening information) features of anxiety. One considerable challenge in assessing and treating anxiety difficulties is that these different response systems are often not closely aligned and may not change in tandem (see seminal work by Lang, Cuthbert, & Bradley [1998] and recent perspective by Taschereau-Dumouchel, Kawato, & Lau [2020], a process termed "desynchrony" or "uncoupling"). Thus, it is important to consider multiple outcome markers to

determine the success of therapy because there is no one *true* measure of fear or anxiety. Analogously, evaluating other markers of impairment in functioning, such as quality of relationships and work life, is critical.

Avoidance of Therapy Itself

Reducing the avoidance of cues associated with panic and phobic targets is typically a key focus in therapy, but avoidance symptoms also have other important treatment implications. In particular, individuals who use avoidance as a coping strategy, albeit a maladaptive one, to reduce their anxiety in the short term also may sometimes desire to avoid treatment and therapy homework, especially if exposure to fear cues is involved in therapy. This may be particularly true in a group treatment context, in which it can be slightly more challenging to permit total flexibility for each individual to set the pace for their exposures. Thus, building a strong therapeutic alliance is essential to help patients become willing to face their fears, and enhancing motivation for treatment may be needed to reduce patients' ambivalence about making changes in their lives, or even about being in treatment (motivational interviewing approaches can be very helpful in this regard; see Miller & Rollnick, 2012). This, of course, would be in conjunction with careful monitoring of patients' avoidance, as well as successful exposure practice between sessions.

Physiological Reactions

A salient characteristic of anxiety disorders is the increased activation of the sympathetic nervous system. Thus, in addition to the subjective feeling of anxiety and the associated behavioral manifestations, there are physiological symptoms such as rapid breathing, increased heart rate, and sweating. These reactions are involuntary and contribute to the feeling of not being in control—a point we return to later. At times, patients are encouraged to tolerate these physiological sensations to learn that they are not actually dangerous, so that the sensations become less frightening; at other times, patients may want to reduce the arousal.

As a means of helping patients reduce their physiological reaction, it can be helpful in some cases to teach them how to use relaxation as a coping skill. This can be accomplished by various methods, most of which involve learning to differentiate between feelings of tension and relaxation, and to use the tension as a signal for reducing their arousal. Using external cues to remind patients to apply the relaxation response can be helpful. Thus, anxious patients can learn to use recurrent situational cues as reminders to relax, such as putting the key in the ignition of the car or sitting down in front of one's computer. Another example is a patient who tended to have panic attacks at restaurants and was able to use the placement of the napkin on her lap as a cue to relax. She was successful in reducing the physiological arousal that was the precursor to panic, and soon she jokingly referred to this as "the magic napkin." Alternatively, patients may be taught to observe their physiological reactions without judging them or attempting to change them, a common approach in more acceptance-based (often called "third wave") therapies, including mindfulness and acceptance and commitment therapy (e.g., Eifert & Forsyth, 2005).

Dysregulation of Multiple Emotions

In some respects, the term "anxiety disorders" is somewhat limiting. While it is certainly true that fear, panic, and anxiety are central emotional states in these disorders, they are not the only forms of affect that become dysregulated or that may require therapeutic attention. For instance, as noted earlier, patients with certain specific phobias often feel a strong disgust reaction to their feared object (see Olatunji & McKay, 2009; Woody & Teachman, 2000). Yet the extent to which it is important to reduce disgust (as well as fear) in treatment remains an open empirical question. While there is some evidence that disgust related to spiders is reduced following successful exposure therapy (e.g., de Jong, Andrea, & Muris, 1997), there is also some indication that disgust may be slower than fear to decline during exposure (e.g., Olatunji, Forsyth, & Cherian, 2007).

Social anxiety disorder is often characterized by feelings of shame and embarrassment, and it is not uncommon for people to feel intense sadness in response to their anxiety disorder symptoms. Along these lines, it is helpful to consider possible interactions between different emotional presentations and traditional gender expectations (e.g., women expressing sadness about their inability to manage their anxiety, and men expressing anger when they feel threatened), though more research is needed to determine the empirical support for these possible gender differences in the treatment context.

Finally, in panic disorder, it is not unusual to see anger as a central aspect of the presentation (Moscovitch, McCabe, Antony, Rocca, & Swinson, 2008), though the tie between anger and depression in these cases needs to be considered. In the case of a panic patient with whom we were unable to simulate panic sensations during the session by means of hyperventilation, we inadvertently triggered a panic attack when he discussed his relationship with his brother. After the panic symptoms had subsided, he reluctantly acknowledged that he was indeed angry at his brother. However, he was raised with a strict religious background and was told at a young age that being angry meant "the devil was in him." Given that he was fearful of feeling or expressing anger, he was first encouraged to acknowledge that he was "frustrated" with his brother, after which he agreed that he was "upset." With further encouragement, he allowed that perhaps he was "annoyed," and finally that he was "angry." Thus, gradual exposure to his concerns about anger became an integral part of the treatment of his panic disorder, as did work on increasing his assertiveness with his brother.

Not Feeling "in Control"

The belief that the world is somehow unpredictable and uncontrollable is central to many anxiety difficulties (Barlow, 2002). Furthermore, one of the clinical hallmarks of panic disorder is that the patients feel "out of control" when panic symptoms arise. In this regard, it is important to frame the therapy intervention as a means by which patients can learn to *manage* their anxiety, so that they no longer feel they are at the mercy of anxiety symptoms and do not find them as threatening. Along these lines, in treatment, we aim to reduce the unhelpful belief that the patient is totally "out

of control" and cannot cope when anxious (e.g., by teaching patients that control is really on a continuum, despite the popular but misleading terminology of being "in" vs. "out" of control). At the same time, we normalize the experience of anxiety and seek to reduce the belief that all anxiety needs to be controlled. Using a coping skills approach to treatment, it is often helpful to tell patients that our goal is not to eliminate all anxiety symptoms, but to help them learn to manage their anxiety by becoming less frightened of anxiety symptoms and their reaction to these symptoms. By understanding that they can tolerate some anxiety and still function, patients actually tend to become less anxious and feel less need to always control their internal (e.g., anxiety symptoms) and external (e.g., rigid behavior) environments.

Limitations of the DSM

It is also important for clinicians to go beyond global diagnostic categories, such as "social anxiety disorder," to adequately assess and treat patients' presenting problems. Specific, idiographic targets of intervention need to be identified (e.g., delineating public speaking concerns or dating fears is more informative than "social anxiety disorder"). Identification of symptoms alone is insufficient, however. A broad biopsychosocial evaluation and case formulation is critical to learn the patient's strengths and weaknesses, and the challenging realities within which some of our patients live.

While improvements in the DSM have clearly been made since it was first published in 1952, it continues to have its shortcomings—particularly for the practicing clinician. The changes have not only helped to bring the classification system more closely in line with the disorders' phenomenology but they also present some challenges in differential diagnosis. For instance, the fact that panic attacks can occur across multiple anxiety problems beyond panic disorder can make it difficult to differentiate whether panic attacks occur in the context of panic disorder, social anxiety disorder, or specific phobia. Imagine a person who regularly has panic attacks while driving; it is possible this person has panic disorder because he fears the consequences of the attack (e.g., that he might faint while driving and cause an accident). Alternatively, it is plausible that this person has a specific driving phobia and has panic attacks tied to his general fears about driving and having an accident. To distinguish among these options, it is helpful to assess whether the attacks also occur in other situations (e.g., when he is not driving), as well as the specific fear triggers and perceived consequences tied to the attacks. Thought experiments in which a patient tries to isolate a given symptom can be helpful in this regard. Consider a woman who has panic attacks when asked to speak in public, and the challenge of determining whether panic disorder and/or social anxiety disorder diagnoses are warranted. Asking the patient how extensively she would still fear public speaking if she could take a magic pill and know that she would not have a panic attack while giving a speech to her coworkers can be useful to determine whether the fears are tied to panic attacks per se, or to fears of negative evaluation. Alternatively, asking how much she would still fear public speaking if there were a risk of having a panic attack but she could take a magic pill, causing her coworkers to think she did a great job, can help to rule in or out fears about the consequences of panic. Thus, *focus* of the fear is important (fear of fainting, bodily sensations, etc., vs. fear of being ridiculed,

appearing stupid, etc.). Not surprisingly, making these fine distinctions in practice can be challenging, especially given that it is quite possible for the patient in this example to have both panic disorder and social anxiety disorder.

Another clinical issue stems from the fact that the social anxiety category is quite heterogeneous, including anxiety in interpersonal interactions, anxiety in public speaking situations, fear of using public bathrooms, and anxiety experienced when signing one's name in public. Moreover, patients who are socially anxious may vary in their social skills, such that behavioral skills training may be appropriate for some but not others (see Rapee & Spence, 2004). Researchers have also found that many gay men who appear to be "socially anxious" are responding appropriately to realistic fears of rejection (Pachankis & Goldfried, 2006); hence, their anxiety may not necessarily be considered unwarranted or maladaptive.

Impairment in Functioning: Love and Work

Given the disruption to relationships caused by avoidance behavior that is characteristic of panic disorder and phobias, and the consequent dependence on loved ones, it is not unusual for families to need some help renegotiating their roles when a patient starts making progress in treatment and is able to function more independently. While this is usually an exciting step, it is not always easy for family members to change long-standing patterns of who are the "strong" or "helper" members of the household. Thus, even with significant progress in achieving anxiety symptom reduction, it is important to evaluate how these life changes influence other aspects of a person's functioning, including possible impairment in interpersonal relationships and work.

We should also emphasize that family members can either assist or interfere with therapeutic progress. For example, researchers found that spousal assistance in the treatment of agoraphobia helped to facilitate the reduction of avoidance behavior (Barlow, O'Brien, & Last, 1984). Serving as a therapeutic aide, the partner of the person with agoraphobia can encourage graduated exposure to feared situations. Sometimes, however, such "assistance" can undercut the therapist's best efforts. One of us (M. R. G.) worked with a woman suffering from panic disorder and agoraphobia, whose husband not only encouraged her to cope with panic by holding her breath but also discouraged her from traveling! The best efforts of the therapist, unfortunately, were no match for this very dominating husband.

COURSE

The conclusions one draws about the typical course and outcomes for persons with anxiety disorders depend in large part on one's focus. Evidence tied to the natural course of these disorders leads to a pessimistic view about their chronic nature, whereas evidence that highlights the tremendous strides in treating these disorders presents a more optimistic picture. At the same time, there is considerable variability in outcomes across disorders, even using similar treatment approaches (e.g., as described later, medication is far more helpful for panic disorder than it is for specific phobias, whereas exposure-based therapies have impressive outcomes for both

specific phobias and panic disorder, with more moderate success rates for treating social anxiety disorder).

Panic Disorder and Agoraphobia

The frequency of panic attacks may wax and wane over time quite dramatically (e.g., a person may have daily attacks for a couple weeks, then far more irregularly spaced attacks over a period of months), though these fluctuations do not seem to occur randomly. The presence of life stressors appears to be a predictor of both the initial onset of panic and its intensity over time (see Klauke, Deckert, Reif, Pauli, & Domschke, 2010). Interestingly, stressors that are tied to either interpersonal difficulties or physical well-being (e.g., scary experiences with drugs) are particularly potent risk factors.

Although the course of panic disorder and agoraphobia may show considerable fluctuations over time, it is difficult to escape the conclusion that the disorders are chronic and disabling. For instance, Nay, Brown, and Roberson-Nay (2013) found that remission rates for panic disorder with and without agoraphobia were 67 and 75%, respectively, at a 3-year follow-up evaluation, with females at a stronger risk for both incidence (odds ratio [OR] = 2.31) and relapse (OR = 2.05) in comparison to males. Unfortunately, even among patients who attain remission, relapse is not unusual (see Batelaan et al., 2010; Bruce et al., 2005). In a naturalistic 5-year follow-up study of clinic outpatients with panic disorder, 38% were labeled as being in remission, 44% experienced mild panic symptoms and phobic avoidance, and 19% had relapsed or were unchanged (Carpiniello et al., 2002). Thus, while many people in this study showed some improvement, residual symptoms were the norm, and relapse was not unusual.

Coinciding with this somewhat discouraging data are the impressive data on treatment outcome, particularly for cognitive-behavioral therapy (CBT) and pharmacotherapy. For instance, a meta-analysis for treatment outcomes for panic disorder (with or without agoraphobia) indicated that CBT combined with pharmacological treatment, CBT alone, and pharmacological treatment alone all showed clear efficacy relative to a variety of control conditions (Bandelow, Seidler-Brandler, Becker, Wedekind, & Rüther, 2007).

CBT can take many forms but typically includes a combination of psychoeducation, exposure therapy, cognitive restructuring, relaxation or slow deep breathing, and relapse prevention (see the widely used manual by Barlow & Craske, 2006). Initially, patients are given psychoeducation about the nature of anxiety and panic, and the "fear-of-fear" cycle (the idea that reacting in catastrophic ways to benign physical sensations makes those sensations more intense and fuels panic attacks). Exposures focus on reducing avoidance behaviors in response to both internal and external panic cues. The exposure assignments typically involve both reentering feared situations (e.g., crowds), while learning to tolerate the anxiety without trying to escape or avoid, and interoceptive exposure exercises, in which patients repeatedly expose themselves to their feared bodily sensations to learn that they can cope with these feelings, and that their anxiety lessens as the feared consequences do not occur. During cognitive restructuring, patients are taught to identify and reevaluate their maladaptive beliefs about the consequences of anxiety and their (in)ability

to tolerate bodily sensations. Also, in some panic treatment protocols, patients are taught diaphragmatic breathing and/or progressive muscle relaxation as methods of coping with stress and anxiety.

The typical length of a CBT panic control treatment protocol, as used in randomized clinical trials, is 10–15 weekly sessions; however, there is evidence that in some instances, positive outcomes can be achieved in much shorter time periods (e.g., five sessions: Otto et al., 2012; two intensive days: Deacon & Abramowitz, 2006). Efficacy has also been found for the administration of computer-based CBT (Andrews, Cuijpers, Craske, McEvoy, & Titov, 2010; Cuijpers et al., 2009). Both individual and group treatments have demonstrated efficacy and are reasonable choices, though outcomes may be slightly enhanced with individual therapy (see review in Craske & Barlow, 2008). Exposure-based therapy for agoraphobia also appears to work well in a group format (Marchand, Roberge, Primiano, & Germain, 2009). In addition to the initial positive outcomes immediately posttreatment (e.g., 65.6% response rate based on the Panic Disorder Severity Scale among patents completing CBT in Aaronson et al., 2008), the available data suggest long-lasting benefits following CBT (e.g., based on 1-year follow-up data; see review by Sánchez-Meca, Rosa-Alcázar, Marín-Martínez, & Gómez-Conesa, 2010), though residual symptoms are certainly not unusual. Furthermore, CBT appears to be a helpful treatment for patients who have not responded to a prior medication treatment (Rodrigues et al., 2011). For more information on empirically supported therapies for panic disorder, see the American Psychological Association Division 12's website on Research-Supported Psychological Treatments (*www.div12.org/diagnosis/panic-disorder*) or the National Institute for Health and Care Excellence's (NICE) guidelines website (*nice.org.uk/guidance/cg113*).

There is also good evidence for the efficacy of medication to manage panic disorder (see summary of pharmacotherapy evidence and guidelines; American Psychiatric Association, 2009). Four classes of medication have demonstrated efficacy: Selective serotonin reuptake inhibitors (SSRIs), serotonin–norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), and benzodiazepines. Although they are relatively comparable in their efficacy, their side effect profiles, risk of dependence, and quality of evidence bases vary considerably. Typically, SSRIs and SNRIs are recommended as a first-line medication treatment because they do not lead to the serious cardiovascular and anticholinergic side effects linked with TCAs, or the high rate of addiction that occurs with benzodiazepines. In fact, there is considerable evidence for long-term negative outcomes with chronic use of high-potency benzodiazepines (see Farach et al., 2012). Thus, benzodiazepine use may be most helpful as part of a short-term addition to one of the other antidepressants to help control symptoms while the antidepressant has a chance to take effect (Susman & Klee, 2005). Even then, the challenge of tolerating the symptoms that often accompany the subsequent discontinuation of benzodiazepines needs to be weighed against the value of the rapid symptom relief they can provide. Fortunately, results for the many trials with SSRIs and SNRIs indicate that there are effective, less addictive medication options available (e.g., Pollack, Mangano, Entsuah, Tzanis, & Simon [2007] found a 70% response rate based on the Panic Disorder Severity Scale among patents completing a venlafaxine ER trial), but discontinuation of these medications requires considerable care (and tapering is often recommended) to help manage possible return of symptoms (see American Psychiatric Association, 2009).

More patients receive medication than CBT as a treatment for panic disorder and agoraphobia, in part because treatment is often sought in primary care settings (where medication is the norm), and in part because CBT is not yet offered in many health care settings despite its established efficacy (see Craske & Barlow, 2008). Any discussion of treatment–outcome for panic disorder and agoraphobia must also consider the typical short- and long-term response of patients receiving a combination of psychotherapy and pharmacotherapy given how frequently this occurs. There are still open questions about when outcomes may be enhanced by combining or sequencing medication and CBT. However, it is clear that combining the treatments is often not necessary and can even be detrimental, despite the seemingly intuitive appeal of using two efficacious approaches (e.g., there is some evidence that medication, and its subsequent withdrawal, can interfere with the long-term benefits of CBT; Barlow, Gorman, Shear, & Woods, 2000; though see Bandelow et al., 2007). The evidence does not seem to suggest that adding CBT to a medication treatment has the same negative impact. For example, adding even single components of CBT to the medication management offered for patients with panic disorder in a primary care setting led to improved outcomes posttreatment and 1 year later (Craske et al., 2005).

The choice to use CBT, medication, or their combination thus depends on the patient's wishes, past treatment history, and so on. However, it is important to highlight that treatment of panic disorder and agoraphobia is one of the field's success stories in terms of positive outcomes in mental health care. As described earlier, not all classes of medications are equal when long-term outcomes are taken into consideration, and caution should be used when combining treatments, as this may be unnecessary and unhelpful in the long run. At the same time, it is nice to have multiple efficacious options from which to choose when offering treatment. This is especially valuable given that not all patients respond to their first treatment, so sequencing treatments for people who do not respond to an initial CBT or medication trial may be a useful way to think about enhancing outcomes with multiple treatment modalities rather than simply combining treatments right away. The issue of using single or combined treatments is also important from a cost-cutting perspective given the tremendous financial burden posed by care for anxiety disorders (e.g., Baxter, Vos, Scott, Ferrari, & Whiteford, 2014). Notably, there is some indication that CBT may be more cost-effective than medication, based on total disability costs, work days missed, health care use, and so on (Heuzenroeder et al., 2004).

While CBT and medication approaches are the primary evidence-based options for treating panic disorder and agoraphobia, there is also some evidence that panic-focused, psychodynamic, twice-weekly, 12-week psychotherapy may also be helpful for treating panic disorder (Beutel et al., 2013; Milrod et al., 2016), though CBT remains a more consistently effective treatment (e.g., Chambless et al., 2017). The psychodynamic panic treatment focuses on confronting the emotional significance attached to panic symptoms, based on the idea that patients are avoidant in part because of difficulty separating from central attachment figures. Also significant is that this intervention explores the role that anger plays in panic attacks, a critical issue, as we noted earlier. It is important to gather further evidence about the efficacy of this approach and its active components. Finally, it is notable that this therapy is still panic-focused and time-limited, and should not be confused with more general supportive psychotherapies that have not been found to be as efficacious as standard panic treatments (Sánchez-Meca et al., 2010).

Therefore, there is both good news and bad news when one examines the course and treatment–outcome for panic disorder and agoraphobia. Studies of the longitudinal course and relatively high relapse rates over time point to the need to develop better long-term management strategies for the disorders, while the impressive CBT and medication efficacy data suggest high treatment response rates. In reconciling these different pictures, a number of factors need to be considered. First, outcomes vary for persons with substantial comorbidity relative to those with relatively "cleaner" presentations of panic disorder. For instance, co-occurring depression is associated with a worse course of illness (e.g., Bruce et al., 2005; Nay et al., 2013), and there is mixed evidence regarding the impact of co-occurring personality disorders (Ansell et al., 2011; Ozkan & Altindag, 2005). Second, many people are simply not receiving treatment, or they are receiving substandard care, which may help explain the discrepant results for the negative, naturalistic course relative to the more positive outcomes for the CBT and medication trials. Finally, panic disorder is undertreated in both primary care (Stein et al., 2011) and mental health care settings (Bruce et al., 2003), with few people receiving CBT or evidence-based medication types and doses (see Wolitzky-Taylor, Zimmerman, Arch, Guzman, & Lagomasino, 2015). Thus, while panic disorder and agoraphobia *can* be treated quite effectively, the field has a long way to go to make this ideal a reality.

Social Anxiety Disorder

Social anxiety disorder tends to be a fairly chronic disorder with relatively low rates of recovery and high rates of recurrence when left untreated. Estimates for the naturalistic course of social anxiety disorder vary widely depending on methodological differences across studies. For instance, retrospective cross-sectional studies indicate that social anxiety disorder persists anywhere between 19 and 30 years in community samples (Kessler, Stein, & Berglund, 1998), with naturalistic recovery rates ranging from 20 to 40% (e.g., see Ruscio et al., 2008). In a prospective study, Bruce and colleagues (2005) found that individuals with social anxiety disorder had only a 37% chance of achieving "recovery" over the course of 12 years (defined as being virtually asymptomatic for at least 2 months). Of those patients with social anxiety disorder who did recover, there was a 39% chance of a recurrence. Taken together, research suggests that social anxiety disorder is an insidious, often unremitting disorder when untreated.

Fortunately, there are a variety of successful psychological interventions available for treating social anxiety disorder, including individual and group CBT. Research generally suggests that combining both cognitive and exposure techniques is recommended (Hofmann & Otto, 2017). Indeed, treatment typically consists of both cognitive restructuring to evaluate beliefs that fuel social anxiety concerns, such as fears of negative evaluation, and exposure exercises to help patients learn to tolerate anxiety and feel more comfortable interacting socially. For instance, cognitive restructuring may target the belief that one will be humiliated by performing "inadequately." Meanwhile, exposures generally focus on encouraging patients to gradually confront and remain in socially threatening situations. This is achieved through the creation of individually tailored fear hierarchies that begin with mildly anxiety-provoking situations and end with the situations that are most feared. When developing hierarchies, it is important to remember that avoidance associated with

social anxiety disorder can take many forms, including not fully engaging with a given situation (e.g., through excessive use of alcohol) or refusing to enter a situation in the first place.

CBTs for social anxiety disorder are demonstrably effective; however, estimates vary considerably in terms of *how* effective. In a review of five meta-analyses evaluating CBT techniques (exposure only; cognitive restructuring only; cognitive restructuring and exposure; social skills training; applied relaxation), Rodebaugh, Holaway, and Heimberg (2004) concluded that each of the CBT interventions revealed moderate to large effect sizes. For more information on empirically supported treatments for social anxiety disorder, see Division 12's website on Research-Supported Psychological Treatments (*www.div12.org/diagnosis/social-anxiety-disorder-and-public-speaking-anxiety*), the NICE guidelines website (*nice.org.uk/guidance/cg159*), or one of the many meta-analyses (e.g., Acarturk, Cuijpers, van Straten, & de Graaf, 2009).

A variety of pharmacological agents have also been used to treat social anxiety disorder. For instance, Mayo-Wilson and colleagues (2014) conducted a meta-analysis to evaluate both psychological and pharmacological treatments for social anxiety disorder. Results indicated that standardized mean differences (*SMD*s) across CBT interventions (*SMD* = –1.19) and pharmacological treatments (*SMD*s ranging from –0.80 to –1.01) were significantly different from the reference comparison, although they did not differ from one another. Of the pharmacological treatments, phenelzine (a monoamine oxidase inhibitor [MAOI]) led to the largest symptom improvements in social anxiety symptoms, but it did not produce significantly better outcomes than other medications included in the investigation (e.g., benzodiazepines, SSRIs, and SNRIs). Given the problematic side effects tied to MAOIs (e.g., risk of hypertension) and benzodiazepines (e.g., physical dependence), medications such as the SSRIs may provide a better first-line pharmacological treatment option for social anxiety disorder (Rodebaugh et al., 2004). (For a more complete list of U.S. Food and Drug Administration [FDA]–approved medications used to treat social anxiety disorder, see *www.drugs.com/condition/social-anxiety-disorder.html*.)

To date, it is not clear whether pharmacological or psychotherapeutic interventions are preferable for treating social anxiety disorder. Some studies have demonstrated that pharmacotherapies may be more effective than CBT in the short term (Blanco et al., 2011), and that the effects of medication on social anxiety may occur more rapidly (Praško et al., 2006). In contrast, there is also evidence that the coping skills learned during CBT may help patients better maintain their treatment gains in the long term (Hollon, Stewart, & Strunk, 2006). Finally, some research indicates no clear differences in the effectiveness of medication and CBT techniques (Roshanaei-Moghaddam et al., 2011). There are also inconsistent findings regarding whether combined psychological and pharmacological treatments for social anxiety disorder are more (or less) effective than either treatment in isolation (see Canton, Scott, & Glue, 2012), leading Turk, Heimberg, and Hope (2001) to recommend that patients with social anxiety disorder stabilize their medication regimens prior to starting therapy and refrain from adjusting or changing their medications during treatment. Treatment decisions likely need to be made on a case-by-case basis, ideally in consultation with both a psychiatrist and a psychologist.

Finally, there is some promising evidence for experimental anxiety reduction interventions (termed "cognitive bias modification") designed to train cognitive

processing biases away from threatening cues, though the ultimate clinical utility of this work remains unknown. For example, Amir, Weber, Beard, Bomyea, and Taylor (2008) found that socially anxious participants trained to attend away from threat-relevant social information experienced decreased social anxiety as assessed by an independent rater. Similarly, Murphy, Hirsch, Mathews, Smith, and Clark (2007) demonstrated that it is possible to experimentally manipulate catastrophic misinterpretations and reduce social anxiety symptoms, even with one brief session of training.

Although meta-analyses make clear the need for strengthening the reliability and effectiveness of cognitive bias modification effects in reducing symptoms (see Cristea, Kok, & Cuijpers, 2015; Hallion & Ruscio, 2011), these experimental interventions raise a number of intriguing theoretical and clinical implications. For instance, in line with cognitive models, this research suggests that not only are cognitive processing biases significantly correlated with symptoms of social anxiety but they are also *causally* related. Additionally, because state anxiety is typically not directly influenced by the training procedures, these studies challenge prior beliefs that it is necessary to activate subjective fear or anxiety directly in order to change elements of the fear network (see Foa & Kozak, 1986) or that verbal mediation is essential to influence cognition and behavior (see Beck & Clark, 1997).

Specific Phobia

Data that address the naturalistic course of specific phobias are relatively sparse, but the general impression is that the disorder in adults tends to be quite stable and enduring without treatment (Becker et al., 2007), although there is variability across phobia subtypes (Depla, ten Have, van Balkom, & de Graaf, 2008). Children with specific phobias, on the other hand, often grow out of their fears (see Kim et al., 2010). For instance, Agras, Chapin, and Oliveau (1972) found that the vast majority of patients under age 20 with specific phobias reported improvement in their fear at a 5-year follow-up assessment. In contrast, fewer than half of adult patients reported improvement.

The picture is quite different when one examines treatment outcome. While individuals with specific phobias rarely present for treatment (McCabe, Ashbaugh, & Antony, 2010), the success rate for exposure-based treatments is very high (e.g., a meta-analysis indicated that the average person receiving treatment was better off than approximately 85% of nontreated persons; Wolitzky-Taylor, Horowitz, Powers, & Telch, 2008).

It is unfortunate that more people do not seek treatment given that improvement can occur rapidly. In fact, there is evidence that a single session of exposure therapy is effective to treat some specific phobias (see Zlomke & Davis, 2008). For specific phobias in general, multi-session (vs. single-session) treatments tend to achieve slightly more favorable results (see Wolitzky-Taylor, Horowitz, Powers, & Telch, 2008), but even these treatments are often brief (e.g., three to eight sessions). Furthermore, the gains made in treatment tend to be maintained for at least 1 year, though more studies with longer follow-up periods are necessary (Choy, Fyer, & Lipsitz, 2007).

The term "exposure therapies" refers to a class of behavioral approaches that are all based on exposing phobic persons to the stimuli that frighten them. The goal is

to provide usually avoidant persons an opportunity to learn that they can tolerate or cope with the fear, which will come down on its own without avoiding or escaping it, and that feared outcomes (e.g., that a snake will lunge, lightning will strike) often do not come true or are not as awful as imagined. The choice of situations to encounter usually follows an individually tailored fear hierarchy that starts with mildly anxiety-provoking situations and builds up to extremely feared situations (though in "flooding" therapy, exposure starts with a very anxiety-provoking stimulus).

Numerous variants of exposure therapy have also been examined, and while many have been found to work effectively, research suggests more favorable outcomes for *in vivo* exposure (which involves actually confronting the feared stimuli) than for other approaches (e.g., systematic desensitization, in which fear-evoking images and thoughts or actual phobic stimuli are paired with relaxation or another response that is incompatible with fear; Choy et al., 2007). Actual contact with feared stimuli also seems to outperform other modes of exposure, such as imaginal or virtual reality exposure (which uses an immersive computer program to generate a phobic situation, e.g., a plane experiencing turbulence). Interestingly, this difference is evident at posttreatment, but *in vivo* exposure's advantage disappears at follow-up, seemingly because those receiving the less direct exposure therapies continue to improve after treatment ends (Wolitzky-Taylor et al., 2008). Another interesting (and effective) variant on standard exposure therapy involves adding applied muscle tension to *in vivo* exposure for the treatment of blood–injection–injury phobia (Ayala, Meuret, & Ritz, 2009). By adding exercises to manipulate muscle tension, patients are able to address the decreases in blood pressure that can lead to fainting with this phobia.

A significant challenge with exposure therapies, especially *in vivo* exposure, concerns fairly high dropout rates and discomfort with the treatment because of the fear associated with trying exposures (Choy et al., 2007). Thus, other treatment options have also been examined, though far less extensively than exposure therapies, and there are some open questions about whether benefits from these treatments that do not include exposure reflect nonspecific treatment effects (due to attention, general support, etc.; see Wolitzky-Taylor et al., 2008). Notwithstanding, cognitive therapies presented either alone or as an augmentation to exposure therapies have received some support, though there is considerable heterogeneity in findings across studies and phobias types (Wolitzky-Taylor et al., 2008).

Finally, the evidence for medication treatment of specific phobias (e.g., use of benzodiazepines or beta-blockers, used either alone or in combination with exposure interventions) has not suggested that anti-anxiety medications consistently add much to treatment outcomes (beyond temporarily making a particular anxiety-provoking encounter, such as a plane flight, easier to endure; see Choy et al., 2007). It should be noted, however, that pharmacological treatments have not been as widely investigated for specific phobias as they have been for other anxiety disorders (Kimmel, Roy-Byrne, & Cowley, 2015). One interesting possible exception to this conclusion about medications concerns D-cycloserine, which is a partial agonist at the *N*-methyl-D-aspartate (NMDA) receptor site. This medication is not designed to reduce fear or the phobia directly; instead, it appears to enhance the pace of exposure therapy by stimulating the amygdala and influencing how readily fear responses are unlearned. While it is too early to determine the ultimate potential for this

medication to enhance treatment of specific phobias, initial evidence (e.g., examining D-cycloserine-enhanced virtual reality exposure for fear of heights; Ressler et al., 2004) has been promising (see Tart et al., 2013).

Clinical Implications

As reviewed earlier, a large and helpful literature evaluates the efficacy of various treatments for panic and phobias. At the same time, this literature leaves open many questions about how to best implement these interventions and what to do when the first-line recommended treatment does not achieve maximal results. Among the clinical implications that follow from this research review are the importance of the therapeutic alliance and clinician's experience, the need to make exposure therapy more appealing to the patient, and the use of general principles of change associated with the treatment of anxiety disorders.

Importance of the Therapeutic Alliance and Experience

We can legitimately give patients good reasons to be hopeful about exposure therapy; the challenge is to get them to try it in the first place and to recognize that tolerating anxiety and exposing themselves to the feared situation, object, or event do not impose any realistic danger. It is here that the therapeutic alliance is most important, in that there needs to be a strong bond between therapist and patient, and both need to be in agreement on the goals and methods of change (Norcross & Lambert, 2019; Norcross & Wampold, 2019). There is no doubt that exposure can effectively reduce avoidance and anxiety—provided that the patient remains in treatment and agrees to engage in the therapeutic procedures. The therapy relationship can increase the likelihood that this will happen.

As we indicated earlier, framing the therapy as a means of teaching coping skills—where patients learn to manage and tolerate their anxiety—is particularly important. There is also some evidence that the experience level of the therapist can play an important role (though this has not been a consistent finding). Mason, Grey, and Veale (2016) found that in providing CBT to individuals with panic disorder and social anxiety disorder, severely anxious clients achieved greater symptom reductions with qualified therapists compared to trainee therapists, though therapist adherence and competence were not measured (see also Huppert et al., 2001). It is not clear whether the difference was a function of the alliance, or the use of more general clinical skills in dealing with issues during the course of treatment, or whether therapist adherence and competence explain these differences.

Making Exposure Therapy More Appealing

In addition to using the therapeutic relationship and gradual exposure (vs. flooding) to make starting the exposure process easier for some patients, the use of relaxation as a coping skill may also help patients to reduce their anxiety and obtain a sense of control. In this regard, however, it is important to note that the literature indicates that some patients (e.g., with panic disorder, generalized anxiety disorder) experience a paradoxical *increase* in anxiety as a consequence of relaxation. Lilienfeld (2007)

has even questioned whether relaxation training is potentially harmful in treating some patients with panic disorder. The phenomenon of relaxation-provoked anxiety in some patients has long been noted in the practice of behavior therapy (e.g., Goldfried & Davison, 1976), and there are indeed ways the skilled clinician can deal with it. Patients who are likely to fear losing control should be forewarned that the sensations they may experience are not signs that they are losing control, but that they are in fact *learning to manage* their sensations of anxiety (see Goldfried & Davison, 1994).

When utilizing relaxation techniques, it is also important to stress that the use of relaxation should not be confused with the inability to tolerate anxiety—an idea that clearly runs counter to the goals of exposure. Accordingly, one of the ways we introduce relaxation exercises in treatment for panic disorder is by noting that relaxation can help to reduce one's resting level of anxiety, making panic and intense anxiety reactions less likely in the first place.

Being Familiar with Principles of Change

In addition to ensuring that clinical interventions are informed by the findings of basic research on anxiety and randomized clinical trials, therapists can enhance their clinical effectiveness by being familiar with general therapeutic principles of change (Eubanks & Goldfried, 2019; Hayes & Hofmann, 2018; Norcross & Goldfried, 2019). Woody and Ollendick (2006) have extrapolated from the research literature a number of principles that can guide practicing therapists in implementing techniques for the treatment of anxiety disorders. Although space limitations prevent us from enumerating all of them, they include general guidelines such as fostering cognitive change, increasing behavioral skills for coping with anxiety and reducing avoidance, and repeatedly using emotionally evocative procedures.

EPIDEMIOLOGY

Examining the prevalence rates for panic disorder and phobias highlights not only how common these problems are but also the fascinating differences in prevalence when comparing rates as a function of gender, age, race, ethnicity and nationality.

Panic Disorder and Agoraphobia

Lifetime prevalence estimates for panic disorder, with or without agoraphobia, range from 3.8 to 4.7%, with 1-year prevalence rate estimates between 1.8 and 2.4% in Western societies (Goodwin et al., 2005, Kessler, Petukhova, Sampson, Zaslavsky, & Wittchen, 2012). The presence of agoraphobia without panic is less common, with an approximately 2.5% lifetime prevalence, and a 1.7% 1-year prevalence (Kessler et al., 2012).

The panic disorder median age of onset is 24 years of age (Kessler et al., 2005), and the onset of panic disorder and agoraphobia appears to be less common in prepubescent children (Kessler et al., 2007), with estimates of less than 0.4% in general pediatric populations (Merikangas et al., 2010). Puberty seems to be an important predictor of panic attacks (even beyond age), and many adolescents do experience panic attacks (Craske et al., 2010).

In general, panic disorder and agoraphobia appear to be less pervasive in elderly populations (Wolitzky-Taylor, Castriotta, Lenze, Stanley, & Craske, 2010). Also, there is some debate about the validity of using DSM criteria with older adults that have been validated primarily in younger adult populations, especially given concerns about high rates of medical comorbidity in older populations in which the symptoms of medical problems may mimic anxiety symptoms (see Mohlman et al., 2012). While panic disorder in older adults typically reflects a preexisting condition (Flint, 1994), a later age of onset certainly does occur. New onset in older adulthood may be related to stressful life events, such as illness or injury, other medical or psychiatric disorders, or emerging dementia (see Bryant, Jackson, & Ames, 2008).

Large gender differences in panic disorder and agoraphobia have been consistently reported (approximately 2:1 more women than men: see Kessler et al., 2006), even cross-culturally (de Jonge et al., 2016). The source of the gender differences is not entirely clear, though some have proposed that sex-based biological differences play a role (see Craske, 2003). Gender roles that indicate it is more acceptable for women to avoid (and for men to turn to substance abuse to self-medicate when they are emotionally dysregulated) may also be contributing factors (see Robinson, Sareen, Cox, & Bolton, 2009). Notwithstanding the gender gap in prevalence, the general course, phenomenology, and treatment response do not vary strongly for men and women, except that women are more likely to have severe agoraphobia (e.g., Bekker & van Mens-Verhulst, 2007).

The data are somewhat mixed with respect to prevalence differences based on ethnic or racial subgroups. A cross-ethnic comparison of lifetime prevalence rates conducted by Collaborative Psychiatric Epidemiology Studies revealed that non-Latinx White Americans consistently endorsed symptoms of panic disorder more than did African Americans, Latinx Americans, and Asian Americans (5.1, 3.8, 4.1, and 2.1%, respectively; Asnaani, Richey, Dimaite, Hinton, & Hofmann, 2010), and that non-Latinx White Americans endorsed more panic symptoms in general, including palpitations, in comparison to African American, Hispanic, and Asian individuals (Asnaani, Gutner, Hinton, & Hofmann, 2009). Other data suggest that Native Americans report significantly higher rates of panic disorder compared to White Americans (Smith et al., 2006), and that African Americans have higher 1-year prevalence rates of agoraphobia compared to White Americans even after researchers controlled for demographic differences (Himle, Baser, Taylor, Campbell, & Jackson, 2009). However, other population-based studies have not found differences in panic disorder and agoraphobia prevalence among African Americans, Hispanic Americans, and European American groups (e.g., Kessler et al., 2006). Notably, there may be some ethnic group differences in the presentation of panic disorder and agoraphobia. For instance, in certain cultures, the triggers for panic attacks are specified by cultural syndromes (e.g., in Latinx groups, shakiness is often considered a disorder of the nervous system and the potential onset of an *ataque de nervios*; Hinton, Chong, Pollack, Barlow, & McNally, 2008). African Americans may have a later age of onset (Lewis-Fernández et al., 2011), and more frequently use coping strategies tied to "counting one's blessings" or religiosity than do European Americans (Friedman & Paradis, 2002). African Americans with panic disorder and agoraphobia also show stronger associations with hypertension (e.g., Friedman & Paradis, 2002) and are more likely to experience panic-like isolated sleep paralysis than do European Americans or other groups with panic disorder and agoraphobia (Paradis & Friedman, 2005).

Epidemiological data across numerous countries suggest comparable lifetime prevalence, age of onset, and gender differences for panic disorder (de Jonge et al., 2016), indicating considerable consistency in the cross-cultural presentation of panic. At the same time, there are some interesting cross-cultural differences in the experience of panic attacks. For instance, in Vietnamese populations, *trúng gió* attacks are relatively common, sharing many symptoms of DSM-defined panic attacks, but the attacks are linked to anticipatory anxiety regarding the pathogenic effect of winds (see Hinton et al., 2007). Khmer refugees in the United States also experience *khyâl* attacks, sometimes called "wind overload," which include common panic symptoms and fear of dizziness upon standing (Hinton, Hinton, Pham, Chau, & Tran, 2003).

Social Anxiety Disorder

Approximately 3–13% of individuals in Western societies have social anxiety disorder at some point in their lifetime (Bandelow & Michaelis, 2015). For instance, the National Comorbidity Survey Replication Study of DSM-IV disorders found that social anxiety disorder has a lifetime prevalence rate of 12.1%, making it one of the most common disorders (Kessler et al., 2005). The median age of onset is 13 years, which is consistent with other research indicating that social anxiety disorder typically develops prior to age 18 (Chavira & Stein, 2005). Along these lines, while social anxiety disorder remains a common disorder in later life, the incidence appears to be lower in older than in younger cohorts. For example, research investigating Canadian adults age 55 and over found that the lifetime prevalence rate for social anxiety disorder in later life was only 4.9% (Cairney et al., 2007).

Similar to many anxiety disorders, in the general population, social anxiety disorder appears to be more prevalent in women than in men (Asher & Aderka, 2018). Interestingly, the sexes are more equally distributed among treatment-seeking populations (Asher, Asnaani, & Aderka. 2017), leading some researchers to speculate that having social anxiety disorder as a man may be relatively more difficult in certain societies (e.g., because of the impairment it can cause in work situations in societies in which men more often tend to be in leadership positions that require public speaking; Rapee, 1995). However, women's pattern of employment and exposure to leadership roles have significantly changed in recent decades, making it important to repeat these evaluations before firm conclusions are drawn. At the same time, research suggests that among individuals with lifetime social anxiety disorder, there are not meaningful gender differences in overall probability of treatment seeking, age at first time of treatment seeking, or delay of treatment seeking after initial onset, but women may be more likely to receive prescriptions for pharmacological treatments (Xu et al., 2012).

When evaluating the epidemiological profile of social anxiety disorder, it is also critical to incorporate cultural considerations. For instance, prevalence rates for social anxiety disorder in Southeast Asian countries (Korea and Taiwan) tend to be much lower than prevalence rates in Western countries (Hofmann, Asnaani, & Hinton, 2010). One explanation for this discrepancy is that the expression of social anxiety disorder may differ cross-culturally. For example, *taijin kyofusho,* a variant of social anxiety common to Japan, is characterized by the fear of embarrassing or causing offense to others. Given the continuous nature of social anxiety symptoms, it is also plausible that the threshold for receiving a diagnosis of social anxiety disorder

differs across cultures, even though symptom profiles are the same (Rapee & Spence, 2004). Finally, it is possible that social interactions may be more "scripted" or regulated in certain cultures, thereby leading to different rates of social anxiety disorder.

Specific Phobia

There is considerable variability in lifetime prevalence rates of specific phobia across studies, with estimates ranging from 6 to 12% (Kessler, Ruscio, Shear, & Wittchen, 2009). Subclinical fears are also extremely common, with the majority of the population reporting a significant fear (Depla et al., 2008). The most frequent phobias in adult samples focus on animals or heights (e.g., Becker et al., 2007), though, again, there is inconsistency across studies.

Collapsing across different types of specific phobias (animal, situational, etc.), the mean age of onset is 8 years old (e.g., Becker et al., 2007), with considerable differences observed across phobia subtypes. For instance, while animal and blood–injection–injury phobias often begin in childhood, situational and height phobias are more likely to emerge in adolescence or adulthood. There are also interesting age differences in overall prevalence rates of specific phobias. Prevalence seems to be highest in childhood and adolescence, peaking at age 21, at which point it follows a fairly stable pattern across the adult lifespan, then decreases around 56 years of age (Oosterink, de Jongh, & Hoogstraten, 2009).

As with the other anxiety disorders reviewed in this chapter, specific phobias are consistently found to be more prevalent among women (e.g., Stinson et al., 2007), especially for fears of snakes, thunder, darkness, and enclosed spaces (Oosterink et al., 2009). Gender differences for phobias of heights, dentists, and flying tend to be lower (see Rowa et al., 2006). There are many possible reasons for the higher phobia prevalence among females, including some intriguing evidence that women are both biologically and socially motivated to avoid threats more than men (see Craske, 2003). Also, more women are apt to seek treatment for specific phobias, so the gender gap is especially pronounced in treatment-seeking samples (McLean, Asnaani, Litz, & Hofmann, 2011).

Research examining variation in the prevalence of specific phobias across diverse cultural, national, and ethnic groups is lacking, and available studies reach conflicting conclusions. For instance, the National Comorbidity Survey Replication found no significant differences in lifetime prevalence of specific phobia across English-speaking Latinx, African American, and non-Latinx White individuals (Himle et al., 2009). Another epidemiological study that included Spanish-speaking Latinx individuals found significantly lower lifetime rates in Asian Americans and Latinx individuals in comparison to non-Latinx White individuals, who did not differ from African Americans and Native Americans (Stinson et al., 2007). The content of phobias also varies by racial/ethnic group, though data are limited. For example, in one study, African Americans endorsed greater animal fears, whereas non-Latinx White individuals endorsed greater blood–injection–injury fears (Chapman, Kertz, Zurlage, & Woodruff-Borden, 2008). Not surprisingly, given variability in cultural norms, taboos, and so forth, across countries, there is also variability in phobia prevalence rates. In one striking example concerning a comparison of 22 countries, phobias were higher in high and higher-middle income countries (8.0–8.1%) than in low/lower-middle income countries (5.7%), with the highest cross-national lifetime

prevalence for animal fears, and the highest differences seen for fear of flying, which had prevalence three times higher in high-income than in low- to lower-middle-income countries (Wardenaar et al., 2017). These data also suggest that individuals with some college or less education compared to those who completed college, and individuals with low and low-average income compared to those with high income, are at higher risk of developing a specific phobia (Wardenaar et al., 2017).

Clinical Implications

Sharing a common diagnostic label is only the starting point in deciding on a treatment plan; cultural and diversity issues need to be carefully considered.

Cultural and Diversity Issues in Assessment and Treatment

In treating any anxiety disorder, it is essential to consider the sociocultural context of patients, such as their ethnic identity, age, gender expectations/roles, sexual orientation, gender identity, and acculturation. These factors should be considered throughout case conceptualization because they will likely play a role in symptom presentation, assessment, and treatment planning. For example, patients may have specific cultural beliefs that panic attacks reflect a punishment from God or an angry spirit. Indeed, there exist some disconcerting examples in which the failure to take into account the unique experiences and beliefs of certain minority groups has resulted in diagnostic bias and inappropriate treatment. Paradis, Friedman, Lazar, Grubea, and Kesselman (1992) found this to be the case in an outpatient clinic for Black patients with panic disorder who had emigrated from the West Indies. The belief system of such patients often includes the practice of voodoo, resulting in an atypical description of symptoms. The authors provide an example of a young woman from Guyana, who came to the clinic describing the typical symptoms of panic that followed a breakup with a boyfriend (e.g., rapid heartbeat, dizziness). In addition to describing the traditional symptoms of panic, she added, "I feel like I'm going to die. . . . I worry it's not natural . . . not natural causes, evil like someone put a curse on me . . . I'm afraid people might look at me and ridicule me because of how I look, people might talk about me" (p. 63). The evaluating clinician gave her the diagnosis "schizophrenia, chronic paranoid type," with the following justification: "The patient's bizarre delusion of an evil spirit inside her body, persecutory delusions of her boyfriend plotting against her, her paranoid delusions of people thinking she is ugly and laughing at her all speak for the diagnosis of schizophrenia, although the onset of delusions at age 33, her normal affect, and the absence of formal thought disorder are inconsistent with the classical presentation of schizophrenia" (p. 64).

Misdiagnoses of social anxiety among gay men similarly fail to take into account the cultural and epidemiological context of the individuals. With the continued presence of verbal and physical harassment of sexual minorities in our society, there are legitimate reasons that such individuals may be concerned about the opinion and reaction of others (e.g., Pachankis, Goldfried, & Ramrattan, 2008).

Finally, when treating older adults, age may be an important consideration throughout case conceptualization. For instance, somatic complaints may mask anxiety problems among elderly patients with late-onset anxiety. Some older adults

choose to describe their symptoms of anxiety as problems with sleep, for example, because this may be perceived as a less stigmatizing way of describing their experiences (Mohlman et al., 2012). Older adults may also be particularly reactive to certain types of anxiety-provoking triggers, such as threats to physical health (Teachman & Gordon, 2009).

COMORBIDITY

Comorbidity among the anxiety disorders is the norm rather than the exception. Interestingly, however, the disorders are by no means uniform with respect to the likelihood or types of co-occurring psychological and medical disorders; thus, we evaluate each disorder in turn.

Panic Disorder and Agoraphobia

Kessler and colleagues (2006) found that 83% of persons with panic had an additional psychiatric disorder. As noted, rates of agoraphobia are very high, with approximately 20% of persons diagnosed with panic disorder in community samples also meeting criteria for agoraphobia, and even higher rates have been observed in clinical settings (Kessler et al., 2006). Comorbidity with other anxiety disorders is also quite common, including generalized anxiety disorder, as well as social anxiety and specific phobias (each occurring in 21–34% of individuals with panic disorder: Kessler et al., 2006).

Mood disorders, both unipolar and bipolar, also frequently co-occur with panic and agoraphobia (Simon et al., 2004). For instance, among individuals with panic disorder, with or without agoraphobia, the lifetime prevalence of major depression is 35–40% (Kessler et al., 2006). Notably, analyses have revealed that panic disorder not only increases the risk for subsequent depression, but that the treatment of panic attacks also may reduce the risk of developing major depressive disorder, suggesting that depression onset may follow the onset of panic disorder (vs. preceding panic; Bittner et al., 2004). Critically, this additional diagnosis of a mood disorder is associated with a worse course for panic, with patients being one-half as likely to recover from panic disorder (Bruce et al., 2005). Although data are limited, there is some indication that depressive symptoms may improve spontaneously following CBT for panic (Allen et al., 2010). All of this raises the question of whether it might be profitable to think of "comorbidity" as reflecting the functional relations among various clinical problems.

Substance use also complicates the presentation of panic disorder, with around 20% of patients with panic disorder meeting criteria at some point for alcohol or other substance abuse (Sansone, Griffith, & Sansone, 2005). In one study, approximately half of those with comorbid panic and substance use disorders developed the panic difficulties prior to the onset of substance use, and 23% indicated use of alcohol or drugs to help reduce anxiety symptoms (Bolton, Cox, Clara, & Sareen, 2006). Dual-diagnosis patients can be especially challenging to treat because of the numerous ways that substance abuse interferes with treatment participation, and there are many open questions about whether integrated or sequential treatment is recommended (see American Psychiatric Association, 2009).

Rates for comorbid personality disorders can also be extremely high, with estimates of 40–50% of persons with panic disorder also meeting criteria for at least one personality disorder; typically, avoidant, obsessive–compulsive, or dependent personality disorder (Grant et al., 2005). This comorbidity has the potential to interfere with treatment when the clinician must contend with pervasive and entrenched personality problems. Complications may be particularly likely if personality difficulties interact with the aims of exposure-based therapy (e.g., building trust and a strong alliance with the therapist in individuals with borderline personality disorder, or encouraging confrontation of fears in persons with avoidant personality disorder).

In addition to these high rates of comorbid psychiatric disorders, medical comorbidity is also a significant issue with panic disorder. Many symptoms of panic attacks share features with the presentation of other, nonpsychiatric medical illnesses, such as myocardial infarction, diabetes, arrhythmia, mitral valve prolapse, hypoglycemia, irritable bowel syndrome, asthma, and chronic obstructive pulmonary disease. This apparent overlap likely contributes to the tendency of people with panic disorder to believe they are experiencing a serious medical illness (Ratcliffe, MacLeod, & Sensky, 2006) and their consequent frequent trips to the emergency room during panic attacks (see Deacon, Lickel, & Abramowitz, 2008). Although panic attacks themselves are not a sign of a serious medical illness, individuals with panic disorder, compared to the general population, do have higher reported prevalence of hypertension, angina, and other cardiac problems; ulcer disease; and thyroid disease, among other illnesses (Meuret, Kroll, & Ritz, 2017).

Social Anxiety Disorder

There are high rates of comorbidity between social anxiety disorder and avoidant personality disorder, with the frequency of avoidant personality disorder among patients with social anxiety disorder ranging from 25 to 100% across a series of studies (see Hummelen, Willberg, Pedersen, & Karterud, 2007). There is also a high rate of comorbidity between social anxiety disorder and both depression and alcohol use disorders. In fact, according to the National Comorbidity Survey Replication, 62.9% of individuals who met criteria for lifetime social anxiety disorder with one to four fears had a coexisting disorder, with percentages even higher for social anxiety disorder with five to seven fears (75.2%), and eight to 10 fears (81.5%). Lifetime social anxiety disorder had a significantly elevated OR for all DSM-IV disorders assessed, including depression (OR = 4.6) and anxiety disorders (OR = 4.9; Ruscio et al., 2008). Meanwhile, 7.2% of individuals with social anxiety disorder met criteria for alcohol abuse, and 9.3% met criteria for alcohol dependence (Chartier, Walker, & Stein, 2003; note that DSM-5 does not delineate between substance abuse and dependence). It is also notable that when social anxiety disorder is comorbid with another psychological disorder, age of onset of social fears has been found to precede age of onset of comorbid disorders in 32% of cases with comorbid anxiety disorders, 71% of cases with comorbid affective disorders, and 80% of cases with comorbid substance dependence/abuse disorder (Chartier et al., 2003). However, this may be an artifact of the early age of onset for social anxiety disorder, as opposed to indicating a causal relationship.

Specific Phobia

Data from the past twenty years suggest that specific phobias often co-occur with other specific phobias; in fact, 75.8% of individuals diagnosed with specific phobia were found to have multiple phobias (Wittchen, Lecrubier, Beesdo, & Nocon, 2003), and many individuals had more than three phobias. In contrast, earlier studies found multiple specific phobias to be relatively rare (e.g., in a Swedish sample, only 18.1% of patients with specific phobias had multiple phobias; Fredrikson, Annas, Fischer, & Wik, 1996), leaving open questions about how measurement issues and cohort effects may be influencing results.

Clinical Implications

Responding to Co-Occurring Symptoms of Physical Illness

It is common for patients with panic disorder to label their physical symptoms as a heart attack. Although this is usually a mislabeling, there nonetheless exists a somewhat higher prevalence of cardiac problems in patients with panic disorder than in the general population. It is therefore important to help patients distinguish between signs of a heart attack and those of a panic attack, and, when relevant, to encourage them to get a full medical examination to determine the presence of a complicating medical condition. When co-occurring medical illnesses are present in the context of CBT for panic, some creativity is often required to adjust exposure exercises so that patients can still meet the goal of eliciting fear and panic sensations while being careful of medically relevant limitations (e.g., hyperventilation exercises can be modified for patients with asthma, so they still get the feeling of shortness of breath).

Alternative Ways to View Comorbidity

As our conceptualization of psychological problems has been shaped over the years by the ascendancy of the DSM, our thinking has changed, so that we understand clinical problems in terms of their diagnostic category. No longer "problems in living," what we see clinically are "disorders." Along with this shift, our understanding of multiple problems is viewed in terms of "comorbidity" rather than the potential functional relationship between problematic emotions, behaviors, and cognitions. The labeling of comorbidity of anxiety and depression in many individuals is quite typical, but unfortunately it does not provide the practicing therapist with guidelines for how to proceed. On the other hand, if one thinks in terms of the functional relationship between anxiety and depression, it is not at all unusual to see functioning impaired because of anxiety and behavioral avoidance, resulting in patients feeling helpless and hopeless about being able to cope with the demands of their lives. For example, patients might receive a diagnosis of both social anxiety disorder and depression due to the fact that they are impaired in both their social and occupational lives by behavioral avoidance, and this failure to function competently leads to depression. Practicing therapists know this to be the case, and it is important for research to focus more on this important clinical issue.

Notably, by focusing on principles of change (e.g., reducing avoidance of distressing situations), rather than diagnoses, it becomes easier to tackle multiple problem

areas simultaneously. Analogously, considering common vulnerability factors that can account for multiple problem manifestations can be helpful in this situation. Notwithstanding, it is often necessary to prioritize targets of therapy because not every problem can be addressed at once. When comorbidity occurs, we generally think about prioritizing or ordering the foci of treatment based on multiple considerations. For instance, is improvement in one domain likely to generalize to other areas of functioning (e.g., will reduced fears of flying likely alleviate depressive symptoms)? What domain does the patient feel is most important to target first? Is one domain versus another (e.g., fear of future panic vs. anhedonia in depression) more clearly impairing functioning? Where are we most likely to effect positive change? Answering these questions is often challenging and may involve some trial and error.

ETIOLOGY

As with most (if not all) psychological disorders, a multitude of factors contributes to the onset of panic disorder and phobias. Thus, isolating the "cause" of a disorder is rarely a realistic goal. Instead, we can talk to patients about the levels of vulnerability and causal factors that may interact to lead to the expression of a specific problem. Barlow (2002) outlines a model of triple vulnerabilities. The first, a *general biological vulnerability,* makes an individual more susceptible to becoming anxious. This first level interacts with a second, *general psychological vulnerability,* which focuses on a sense that the environment is uncontrollable and unpredictable. The combination of these two broad vulnerabilities can lead an individual to become extremely fearful in situations that are not objectively dangerous (termed "false alarms"). The third level constitutes *specific psychological vulnerabilities* that lead to the expression of an identifiable anxiety disorder as a function of an individual's particular learning experiences about what aspects of the environment are threatening. For instance, learning that evaluation by others can be dangerous may lead to social anxiety disorder, and learning that heights are hazardous may result in a specific phobia, whereas learning to fear changes in bodily sensations may manifest as panic disorder. Given the considerable overlap in etiological models, we highlight the common vulnerabilities for panic disorder and phobias in this section. However, we also provide examples of the unique ways in which these vulnerability factors manifest across the specific disorders.

General Biological Vulnerabilities

At the biological level, family and genetic studies point to the conclusion that there is a broad genetic basis to anxiety and related areas of emotion dysregulation (e.g., depression). For example, first-degree relatives of samples with anxiety disorders have higher rates of anxiety difficulties than the relatives of nonanxious control samples (e.g., Weissman, 1993). Similarly, the children of parents with anxiety disorders have been observed to have more than nine times greater risk of having a psychiatric disorder than the children of nonanxious control parents (see Beidel & Turner, 1997). More specifically, the children of parents with panic disorder tend to be more anxious than the children of nonanxious control parents (e.g., Sylvester,

Hyde, & Reichler, 1988). These children also tend to be more depressed, pointing to the general nature of the vulnerability (though some anxiety disorders may have a more disorder-specific transmission, such as specific phobias; Page & Martin, 1998). Along these lines, genetic factors appear to play a moderate but meaningful role in the development of social anxiety disorder, explaining approximately 30–50% of the variance in the disorder (Kendler, Karkowski, & Prescott, 1999). However, it is less clear whether these genetic factors are specific to the development of social anxiety disorder, as opposed to the development of mental health problems more broadly (Kimbrel, 2008). Finally, twin studies, which help to minimize potential confounds between genetic and nongenetic influences, also support the heritability of a general anxiety proneness. For instance, there are higher concordance rates of anxiety disorders in monozygotic (12–26%) than in dizygotic (4–15%) twins (Hettema, Neale, & Kendler, 2001). Critically, the evidence to date does not point to a specific anxiety gene; instead, there appear to be relatively weak contributions from an array of genes (Smoller, Block, & Young, 2009). It is also important to highlight that many of the studies on biological vulnerabilities do not guarantee that the finding of higher rates of anxiety disorders among family members is due purely to genetic factors, as opposed to other influences, such as social learning experiences.

Temperamental vulnerabilities tied to anxiety (Kampman, Viikki, & Leinonen, 2017) also seem to play a role in setting the stage for later expression of an anxiety problem. For example, behavioral inhibition, "a temperamental trait observed in infants and young children characterized by behavioral withdrawal, decreased approach behavior, increased vigilance, and increased arousal in response to novel and unfamiliar situations" (Kimbrel, 2008, p. 594), has been found to predict later anxiety disorders. Persons with a stable pattern of high behavioral inhibition were more likely to develop an anxiety disorder, especially social anxiety disorder (Fox, Henderson, Marshall, Nichols, & Ghera, 2005), and behavioral inhibition at age 8 was linked to increased risk for anxiety and depressive disorders at ages 16–21 (Goodwin, Fergusson, & Horwood, 2004). Behavioral inhibition has received considerable attention as an early risk factor within the context of social anxiety disorder. For example, Clauss and Blackford (2012) found that 43% of behaviorally inhibited children met criteria for social anxiety disorder as compared to 12% of noninhibited children.

Numerous other biomarkers have also been linked to heightened risk for anxiety disorders, though the strength of the evidence varies considerably across markers. For instance, an imbalance of the neurotransmitter serotonin has often been proposed as a factor in the etiology of anxiety pathology (e.g., in panic disorder; Stein & Hugo, 2002), but this proposal is based mostly on the effective treatment response of anxiety to SSRIs rather than on prospective evidence. Other proposed neuropsychiatric contributions, derived in part from animal models, include differences in the anatomy of the amygdala and hippocampus, and altered stress responses with respect to cortisol levels and the hypothalamic–pituitary–adrenal axis (see review by Uhde & Singareddy, 2002).

There is also considerable evidence that certain biological triggers make panic attacks more likely, and there is a rich tradition of eliciting panic attacks in the laboratory by altering respiration in vulnerable individuals (e.g., through inducing hyperventilation or inhaling CO_2; see Barlow, 2002, Chapter 5). This has led to the suggestion that individuals who are particularly sensitive to increased levels of

CO_2 may be at greater risk of developing panic attacks, an idea known as the "false-suffocation alarm theory" (Klein, 1993), though this hypothesis has received only mixed empirical support (see Barlow, 2002). Notably, the likelihood of cueing a panic attack through a biochemical or respiration challenge depends in part on other biological vulnerabilities (e.g., respiratory hypersensitivity; Blechert, Wilhelm, Meuret, Wilhelm, & Roth, 2010). Furthermore, the occurrence of panic attacks following a biological challenge is heavily influenced by other psychological vulnerabilities, such as greater baseline experiential avoidance (e.g., Karekla, Forsyth, & Kelly, 2004) and anxiety sensitivity (e.g., Blechert, Wilhelm, Meuret, Wilhelm, & Roth, 2013), which brings us to the second level of vulnerabilities.

General Psychological Vulnerabilities

Interacting with general biological vulnerabilities is a set of broad psychological vulnerabilities that set the individual up to see the world as an uncontrollable and unpredictable place. This focus on the importance of perceived and actual control is based partly on extensive evidence from animal models that show having minimal control over the environment during early development can lead to chronic difficulties with anxiety (e.g., work with rhesus monkeys; Suomi, 2000). In humans, an attributional style reflecting a sense that outcomes are uncontrollable, either in general or because of personal limitations, also seems to be a risk factor (see Reardon & Williams, 2007). Interestingly, this cognitive style may be more common in girls, which may contribute to the often-cited gender differences in anxiety disorder prevalence rates. Hilt and Nolen-Hoeksema (2009) propose that this gender disparity may be due to higher rates of negative life events during childhood and adolescence among girls (e.g., sexual abuse), and more experiences that suggest one's behavior has limited impact on the environment. Other evidence that early learning experiences about controllability and predictability are important follow from the finding that parenting styles promoting a sense of control by being contingently responsive yet not overprotective (i.e., permitting exploration) seem to protect against the development of anxiety problems (see Murray, Creswell, & Cooper, 2009). It is important to note that most of the evidence in humans linking uncontrollability with anxiety is correlational given the ethical barriers to manipulating children's environments to enhance uncontrollability.

In addition to early learning experiences that focus on uncontrollability, stressful life events more generally seem to enhance vulnerability. These events serve as triggers for various forms of psychopathology and seem most closely implicated in anxiety pathology if the events are either threatening (e.g., involvement in an accident) or uncontrollable in some way (e.g., Eley & Stevenson, 2000). For instance, patients with panic disorder reported a higher number of salient life events, especially those that seemed out of their control in the 12 months before onset of panic relative to the same time period for control participants (Klauke et al., 2010).

The interpretation of an event as negative and stressful is, of course, highly dependent on how individuals process that event, and there is extensive evidence that anxious individuals selectively process threatening information (see MacLeod & Mathews, 2012). These cognitive biases are thought to maintain anxiety and dysfunctional avoidance behaviors by keeping threatening cues salient to the vulnerable

person. For instance, the general tendency among anxious samples to interpret ambiguous situations as threatening manifests as a focus on (mis)interpretations of bodily sensations in panic disorder. Specifically, the cognitive model of panic attacks was developed in part from observations of pharmacological and neurochemical studies of agents that promoted panic attacks, but only in select individuals—those who tended to interpret the bodily sensations induced by the agents in a disastrous way (Clark, 1986). This led Clark to suggest that panic attacks occur because certain bodily sensations are misinterpreted as indicating a catastrophe, such as a heart attack or loss of control. In social anxiety disorder, cognitive processing biases and related emotional responses in social situations are thought to reinforce the idea that social or performance situations are threatening, and that negative evaluation is likely (Heimberg & Magee, 2014). When conversing with acquaintances at a party, for instance, individuals with social anxiety disorder may notice the one person who does not smile at something they have said and interpret this as evidence that they are uninteresting.

In addition to the abundant data indicating that selective information processing is associated with anxiety disorders (see MacLeod & Mathews, 2012), there is also evidence that these information-processing biases enhance vulnerability to react more negatively to subsequent challenging life events (e.g., MacLeod, Rutherford, Campbell, Ebsworthy, & Holker, 2002). Similarly, "anxiety sensitivity" (the belief that anxiety symptoms are themselves scary) and explicit threat appraisals have been found to predict the onset of anxiety disorder symptoms (e.g., Schmidt, Zvolensky, & Maner, 2006).

Specific Psychological Vulnerabilities

The combination of general biological and psychological vulnerabilities is thought to enhance anxiety proneness, while an individual's particular learning experiences are expected to predict whether a specific type of anxiety disorder is expressed. For instance, the emergence of panic disorder would be especially likely if the dangers of somatic symptoms and illnesses were emphasized and reinforced during childhood (see Barlow, 2002). Meanwhile, "sensitizing social experiences," such as being excessively teased, are thought potentially to increase the risk for developing generalized social anxiety disorder (Kimbrel, 2008).

There is also considerable evidence that conditioning experiences can be central to the onset of phobias (see Field, 2006), and stories of a traumatic dog bite leading to a dog phobia are not unusual. Furthermore, Mowrer's (1947) influential two-factor theory includes both classical and operant conditioning mechanisms—specifically, fear becomes a classically conditioned response to a previously neutral stimulus through a negative conditioning event, then avoidance and escape behaviors are operantly reinforced because of their (temporary) anxiety-reducing qualities. However, classical conditioning models are not able to explain all the varied ways that fear acquisition can occur (i.e., lots of people who experience dog bites do not go on to develop a phobia and many, if not most, people with phobias did not experience a bad bite). Thus, Rachman (1990) has suggested multiple pathways to acquire specific fears. In addition to classical conditioning, a person may vicariously learn via modeling (e.g., watching someone else be bitten by a dog), or simply through the

transmission of information that a given situation is dangerous (e.g., reading about plane crashes; see also Mineka & Zinbarg, 2006).

An additional challenge in explaining the development of fear and phobias concerns the unequal distribution of fears in the population (e.g., particularly high rates of snake and spider phobias relative to other animal phobias, despite fewer stories of snake bites than dog bites). This has led to the suggestion that we are predisposed to acquire some fears very readily because it was adaptive to be fearful of certain objects or situations in our evolutionary past. This idea, termed "biological preparedness theory" (Seligman, 1971), has some support from nonhuman primate studies (e.g., monkeys are more likely to exhibit fear when observing another monkey showing fear in response to a snake than to a rabbit; Cook & Mineka, 1989), though the broader status of preparedness theory remains controversial (see McNally, 2016).

Linked to early learning experiences, social skills and behaviors also feature prominently in many etiological models of social anxiety disorder. For example, Rapee and Spence (2004) draw a distinction between poor social skills and "interrupted social performance," which they define as "the interference of appropriate social behavior due to heightened anxiety" (p. 758). According to Rapee and Spence, interrupted social performance likely plays a larger role in the maintenance (vs. development) of social anxiety. At extreme levels, however, interrupted social performance may also interfere with one's ability to obtain and nurture meaningful friendships, which in turn could contribute to the development of social anxiety disorder.

It seems clear at this point that it is the interaction of a range of general and specific biological and psychological vulnerabilities that explains why a given person develops an anxiety problem. Even in the primate studies, the likelihood of fear conditioning is dependent in part on the monkeys' prior experience. For instance, fear conditioning is less likely if the monkeys have previously observed nonfearful behavior in response to the potential fear stimulus (Mineka & Cook, 1986). Thus, updated learning models (e.g., Mineka & Zinbarg, 2006) tend to be more comprehensive and reflect the likely interaction of biological, cultural, and environmental vulnerabilities, as well as particular stressors that can influence the impact of a given learning experience.

Clinical Implications

Here, we consider the question of connecting etiology to treatment, and the diversity of developmental pathways for a given anxiety disorder.

Connecting Etiology to Treatment

When behavioral approaches to treatment were first introduced in the late 1950s, it was proposed that knowledge of etiology was not needed to plan treatment. It was assumed that problematic behavior—especially anxiety—was learned, and what the patient needed was new learning experiences. Consequently, intervention procedures were developed that focused on the current life situation, and not the past. Although contemporary CBT continues to emphasize the here and now, a focus on etiology can have important therapeutic benefits, especially in discussing the ongoing contributions to current fears from early life experiences. Take the example of

socially anxious individuals who were frequently berated by others for being shy. While their anxiety may be partially a function of a biological vulnerability, a psychoeducational intervention could potentially allow them to reattribute one of the "causes" of their problem to something other than themselves.

Diverse Developmental Pathways for a Given Disorder

Given that the DSM classification system is topographical and does not assess the function associated with the symptoms for a given disorder, it can include within a single category clinical problems that have different developmental pathways. Thus, one individual who is anxious in social situations may have adequate social skills but nevertheless be overly concerned about the negative reaction of others. Meanwhile, another individual may have serious deficiencies in their social skills, so their concern about the negative reaction of others would be more realistic.

The etiological factors underlying panic can be highly idiosyncratic. For example, one of us (M. R. G.) worked with an actress whose panic disorder caused her considerable anxiety prior to stage performances. In conducting an assessment of the trigger for her initial panic attack, the therapist inquired whether she had recently undergone any surgical procedures that might have been stressful. The actress indicated that she indeed had a "tummy tuck" shortly before her initial panic attack. As the therapy progressed and as abdominal breathing was introduced as one of the interventions, the patient indicated that she already knew how to do this and had used this successfully throughout her career. However, when she was asked to demonstrate her abdominal breathing ability, it was apparent that she was not doing it properly; indeed, the tummy tuck interfered with her ability to engage in abdominal breathing, likely enhancing her anxiety and setting her up to panic when she practiced this technique before going on stage.

CONCLUSION

Our goal in this chapter has been to summarize the basic and applied research on panic disorder, social anxiety, and phobias, and to extrapolate clinical implications from these findings to aid therapists in developing their case formulations and treatment plans. In our review of the research literature on treatment, our focus has been primarily on the cognitive-behavioral techniques that are demonstrably efficacious. However, we also noted the very critical role of the therapeutic relationship, acceptance of symptoms, agreement between therapist and patient on the goals for treatment, and an agreement about the specific intervention procedures. Furthermore, treatment is not one size fits all. Practicing clinicians have always known this to be the case, but it appears that the research literature needs to focus more on this concept. For instance, research focused on cultural considerations, treatment effectiveness in real-world settings, and therapeutic outcomes for patients with dual diagnoses is sorely needed.

Although there are specific procedures for particular clinical problems, it is also important for the clinician to keep in mind the overriding principles of change that may be reflected in the specific procedures. Among these are (1) a positive

expectation that therapy will help and the presence of sufficient motivation to participate; (2) an optimal therapeutic alliance; (3) doing what is necessary to help patients to become better aware of what they are doing and not doing, and how that contributes to their life problems and symptoms; (4) corrective experiences (e.g., via exposure) that encourage new learning and result in altered emotions and behavior, followed by changes in expectations that encourage further corrective experiences; and (5) reevaluation of biased thinking (Eubanks & Goldfried, 2019).

Human behavior is far too varied and complex to be understood by any single therapeutic model or procedure. If we think in terms of general principles of change and routine monitoring of therapy progress and outcomes, we can then begin to investigate the parameters associated with an intervention's efficacy and effectiveness for different patients. In pursuing such an evidenced-based approach, we can begin to build a body of knowledge that—rather than being restricted by any single theoretical orientation—is empirically based.

Finally, while this chapter is structured to derive clinical implications from research findings, the clinical situation should also be thought of as the context of discovery, whereby therapists' clinical observations provide invaluable directions for hypothesis generation and future research. To have a meaningful interaction between research and practice, the bridge needs to go both ways. After a drug has been approved by the FDA following randomized clinical trials, practitioners have the opportunity to provide feedback to the FDA on the use of the drug in clinical practice. The Society of Clinical Psychology, Division 12 of the American Psychological Association, established a similar mechanism, providing a two-way bridge between research findings on empirically supported treatments and therapy experiences that use these treatments in practice (*www.stonybrook.edu/commcms/two-way-bridge*).

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CHAPTER 5

Obsessive–Compulsive Disorder

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Lynne Siqueland

Our goal in this chapter is to outline and discuss the psychopathology of obsessive–compulsive disorder (OCD), especially as it guides implementation of treatment of this often baffling disorder.

OCD SYMPTOM PRESENTATION AND CLASSIFICATION

OCD is characterized by two main symptoms: obsessions and compulsions (American Psychiatric Association, 2013). "Obsessions" are unwanted thoughts, ideas, images, or impulses and urges that are experienced as repugnant, threatening, obscene, blasphemous, nonsensical, or all of these. The content of obsessions is highly individualized and shaped by the thinker's personal experiences, although the general themes of obsessions can be organized into several general categories: contamination, responsibility for harm, violence, sex, religion, and the need for order or exactness (e.g., Abramowitz et al., 2010; McKay et al., 2004). Most people with OCD evince multiple obsessional themes and forms, and the form and content of these may shift over time. Table 5.1 provides examples of common obsessions. Obsessions often evoke intense distress and impairment that result from avoidance of related situations. For example, someone with obsessions about harming hurting or killing family members (e.g., spouse, child) may avoid loved ones and experience depressive symptoms as a result.

Three characteristics set clinical obsessions apart from other types of repetitive thinking (e.g., worries, daydreams). First, obsessions are experienced as *unwanted* or *uncontrollable,* in that they *intrude* into consciousness (often triggered by cues or something in the environment). Second, while personally relevant, the content of

TABLE 5.1. Common Obsessions

Category	Examples
Contamination	• Concerns about getting AIDS from touching a bathroom door knob • Fear of getting cancer from using lawn fertilizer • Fear that "germs" will spread to others and make them ill • Fear of getting sick from touching dirty laundry
Responsibility for harm	• Thought that you've hit someone with your car without realizing it • Doubts about whether the door is locked or the oven is turned off • Doubts about whether you dropped pills on the floor and a child thought they were candy and ate them
Symmetry/order	• The need to have the environment "balanced" on the left and right sides • Thoughts that a parent will die if the books aren't arranged in alphabetical order
Sexual	• Impulses to kiss one's boss • Thoughts of having sex with children or family members • Doubts regarding whether one is homosexual or heterosexual
Violence/aggression	• Impulses to harm loved ones • Thoughts of injuring a vulnerable person (small child or elderly)
Religious	• Blasphemous images of Jesus with an erection on the cross • Doubts about whether one is "faithful enough"

obsessions is *incongruent* with the individual's belief system and is not the type of thought one would normally expect oneself to have. Third, obsessions are *resisted*; that is, they are accompanied by the sense that a fix is needed, so the obsession must be "dealt with," neutralized, or altogether avoided. The motivation to resist is activated by the fear that if action is not taken, disastrous consequences may occur.

Compulsive rituals are the most noticeable feature of OCD and often account for the most functional impairment. The DSM specifies that "compulsions" are motivated and intentional, in contrast to mechanical or robotic repetitive behaviors, as observed in disorders such as Tourette syndrome. Moreover, compulsions are performed in response to obsessions and to reduce obsession-related distress. This is in contrast to repetitive behaviors in addictive or impulse control disorders problems (e.g., "sexual addiction," trichotillomania), which are carried out because they may produce pleasure or gratification. Examples of the most common compulsive rituals include washing and cleaning, checking, seeking reassurance, ordering and arranging, counting, and repeating routine behaviors (e.g., stepping through a doorway, flicking a light switch, getting up from a chair). Specific examples of common rituals are listed in Table 5.2.

Studies on the structure of OCD symptoms consistently identify *symptom dimensions* that comprise certain obsessions and compulsions, including (1) harming (aggressive obsessions and checking rituals), (2) contamination (contamination

obsessions and decontamination rituals), (3) symmetry (obsessions about order or neatness and arranging rituals), and (4) unacceptable immoral or violent thoughts with mental rituals and neutralization (e.g., Abramowitz et al., 2010). These symptom dimensions overlap, such that someone might have, for example, both contamination and harming symptoms. In addition, some sufferers undergo changes in their symptom picture over time with changes in both obsessions or compulsions, for example, shifting from unacceptable thoughts to symmetry concerns. Each of these symptom dimensions also has implications for the psychological treatment of OCD because the use (and effectiveness) of specific treatment techniques depends in part on the patient's symptom presentation (see treatment section).

In many instances, it is clear how compulsive rituals are intended to reduce obsessional anxiety about particular feared consequences—for example, compulsively checking the stove/door locks to reduce the fears of a fire/break-in, or repetitive handwashing rituals intended to remove feared contaminants and prevent disease. In some cases, patients perform rituals simply to reduce a general sense of anxiety or the sense of "incompleteness." Compulsive rituals can also be overt behaviors or covert mental acts aimed at preventing a negative outcome. Examples of common mental rituals include repetition of special phrases, prayers, or numbers in a specific manner, and ritualistically going over (mentally reviewing, analyzing) one's behavior or conversations to reassure oneself that one has not made egregious mistakes or said anything offensive.

In addition to compulsive rituals, many people with OCD use various overt and covert strategies that do not meet DSM-5 criteria to control, remove, or prevent their obsessions (Abramowitz & Jacoby, 2014; Ladouceur et al., 2000). Examples include overanalyzing the meaning of obsessional thoughts, rational self-talk (i.e., to convince oneself of the unimportance of the obsession), reassurance seeking, replacing the obsessional thought with a "safe" thought, performing a brief behavioral act (e.g., putting one's hand in one's pockets to prevent losing control), distraction, and

TABLE 5.2. Common Compulsive Rituals

Category	Examples
Decontamination	• Handwashing for 15 minutes at a time • Ritualized showering routines, including washing the shower itself • Wiping down mail and groceries before bringing them into the house • Demanding that others wash before touching items in the house
Checking	• Locks, windows, appliances, paperwork • That others are safe • Asking repeated questions of others to gain assurances
Repeating	• Going up and down the stairs several times until an unwanted thought is dismissed • Getting dressed and undressed over and over, until it feels "just right"
Mental rituals	• Repeating prayers, words, or phrases to oneself to neutralize unwanted thoughts • Mentally reviewing what one said or did (reassurance)

thought suppression (Ladouceur et al., 2000). Because these strategies may be covert and automatic, the individual may not even recognize these behaviors as part of OCD. Thus, the assessment should include examples to help the individual identify neutralization.

Avoidance is also present to some degree in most individuals with OCD, and is intended to prevent exposure to situations that provoke obsessional thoughts and urges to ritualize. Sometimes the aim of avoidance is to prevent specific feared consequences, such as illness or harm. Other times, avoidance is used to prevent obsessional thoughts, such as avoiding public stairways because they evoke obsessions of impulsively pushing people down the steps.

Differentiating OCD from Similar Disorders

Illness Anxiety Disorder and Body Dysmorphic Disorder

Illness anxiety disorder (IAD; formerly "hypochondriasis") is a somatic symptom disorder characterized by a preoccupation with fears of having a serious disease (e.g., cancer). The intrusive health-related preoccupations in IAD have been compared to obsessional thoughts or fears in OCD, and the repetitive attempts to seek reassurance in IAD, to compulsive checking rituals (e.g., Fallon, Javitch, Hollander, & Liebowitz, 1991).

Body dysmorphic disorder (BDD), which is categorized as an OCD-related disorder in DSM-5, involves extreme concerns about one or more physical flaws that either are not real or are so slight that others would not notice (e.g., "My nose is too large").

The thoughts of unsightliness occur in the absence of any noticeable physical defect, are often resisted, and lead to significant anxiety about how the person appears to others. In this way, preoccupations in BDD are functionally similar to obsessions in OCD. To circumvent anticipated embarrassment, individuals with BDD may avoid certain social situations or engage in behaviors aimed to reduce distress or the visibility of their imagined defect, including excessive checking in mirrors, grooming, and hiding the perceived defect (e.g., with large sunglasses or a hat). Because IAD and BDD involve anxiety-provoking preoccupations and excessive behaviors aimed at reducing this anxiety, these problems might be related to OCD. In clinically distinguishing IAD and BDD from OCD, it is important to assess the full range of obsessions and compulsions, and determine whether any health and appearance concerns occur along with other more quintessential types of obsessions and compulsions. If this is the case, OCD is the proper diagnosis. If the person appears singly obsessed with appearance or health concerns, somatic symptom disorder is likely the correct diagnosis.

Generalized Anxiety Disorder

The main features of *generalized anxiety disorder (GAD)* are chronic worry and tension that are unfounded or more severe than would be expected in most people, and that lead to insomnia and other physical symptoms, including fatigue, headaches, irritability, and hot flashes. Worries in GAD can be intrusive, unwanted, repetitive, and highly distressing to the individual; thus, they are often mistaken for obsessions. Yet

whereas the content of worries in GAD is focused on real-life circumstances such as finances, social and family relationships, work, and school performance, obsessional content in OCD does not typically concern everyday problems. Worries also tend to involve exclusively verbal content, whereas obsessions are often experienced as images and impulses, along with thoughts and doubts. Worry content often shifts from one topic to another over time, whereas the focus of obsessions is typically stable. Some individuals with GAD show ritual-like reassurance-seeking and checking behavior (e.g., calling one's spouse repeatedly if one is late getting home); however, these behaviors are more specific to worry content and do not neutralize anxiety in the same way that compulsive rituals neutralize obsessions.

Obsessive–Compulsive Personality Disorder

The main features of *obsessive–compulsive personality disorder (OCPD)* are an enduring pattern of perfectionism, rigidity, stubbornness, and orderliness that interferes with task completion; preoccupation with rules, organization, and schedules, so that the point of activities is lost; overconscientiousness and inflexibility regarding ethical or moral issues (not accounted for by normal cultural or religious values); and excessive devotion to work and productivity to the exclusion of friendships or leisure time (American Psychiatric Association, 2013). Although some of these characteristics are informally referred to as "compulsive" and might be found among some individuals with OCD, there are important differences in the *function* of these symptoms. In OCD, perfectionistic behavior is provoked by fear and is resisted; the person wishes they did not feel compelled to behave this way. In contrast, people with OCPD do not have obsessional fears and instead perceive their "compulsive" traits as useful, agreeable, and consistent with their worldview. Thus, when individuals do not resist their behavior patterns and reject the suggestion of adapting a less rigid lifestyle, OCPD is the likely diagnosis rather than OCD.

Schizophrenia and Delusional Disorders

It is important to distinguish schizophrenia and delusional disorders from OCD, since these conditions possess superficial similarities that have led some to speculate a relationship among them (e.g., Enright, 1996). For instance, these other conditions may involve bizarre thoughts (i.e., "delusions") and behavior. Yet unlike in OCD, most people with schizophrenia and other psychotic disorders do not resist their bizarre, intrusive thoughts, and the thoughts do not produce anxiety or urges to neutralize. Rather, the bizarre ideation might be believed or even defended as reasonable (i.e., as in delusions). Other differences are that repetitive behaviors in schizophrenia (e.g., undifferentiated or catatonic subtypes) and other psychotic disorders are pointless motor behaviors rather than responses to obsessional anxiety. Nevertheless, patients with OCD are sometimes misdiagnosed as having delusions or psychotic thinking, especially if the OCD is characterized by bizarre obsessions (e.g., fear of making a pact with the devil, intrusions about murdering loved ones). Therefore, the unwanted nature of the thoughts and the anxiety response, in concert with the presence of avoidance and compulsions (overt and covert), must be carefully assessed, as these features differentiate psychotic symptoms from those of OCD.

Clinical Implications for Understanding the Role of Obsessions and Compulsions

The clinician should keep in mind that obsessions and rituals (including avoidance and neutralizing) are *functionally related*—the compulsive ritual is an attempt to reduce obsessional distress. For example, the obsession that one has mistakenly hit a pedestrian while driving would *increase* anxiety and lead to the urge compulsively to check (ritual) or ask for assurances to *reduce* this anxiety. It is therefore not surprising that certain types of obsessions and rituals co-occur in the following patterns: (1) Obsessions about responsibility for harm and mistakes are associated with checking rituals; (2) symmetry obsessions occur along with ordering and counting rituals; (3) contamination obsessions co-occur with washing and cleaning rituals; and (4) religious, sexual, and violent obsessions often trigger mental rituals and reassurance seeking (e.g., McKay et al., 2004).

It is also important for the clinician to understand the content and triggers of obsessional thoughts, as well as the feared consequences predicted to result from the obsessions. Information about the compulsive rituals and avoidance strategies must be collected, including the individual's rationale for deploying each type of ritual; that is, how does the ritual prevent feared negative consequences? This understanding of the function of the symptoms and the relationship between obsession and compulsion is essential to accurate diagnosis of OCD and its lack leads to frequent misdiagnosis and treatment recommendations. The rationale and effectiveness of psychological treatments for OCD hinge on this conceptualization. Finally, it is important to determine individuals' insight into the senselessness of their symptoms, since poor insight is associated with attenuated outcome (e.g., Foa, Abramowitz, Franklin, & Kozak, 1999).

COURSE AND ASSOCIATED FEATURES

OCD is a chronic condition with a very low rate of spontaneous remission. Left untreated, symptoms fluctuate, with worsening during periods of increased life stress. In a 40-year follow-up study of 144 individuals with OCD (Skoog & Skoog, 1999), 83% of this cohort had improved, and 48% no longer met diagnostic criteria for OCD, although about half of the nonclinical individuals reported some residual symptoms. Thus, although OCD symptoms are likely to improve with treatment, full recovery is the exception rather than the rule. The clinical implications of this fact are discussed in the section on treatment.

Having OCD is associated with impaired social and role functioning, troubled romantic and family relationships, diminished academic performance, unemployment, comorbid psychological disorders, sleep and sexual problems, and increased receipt of disability income (Koran, 2000). Despite the fact that severe OCD can lead to functional impairment, people with OCD do not differ substantially from the general U.S. population in terms of the rates of alcohol abuse, suicide, or marriage. Relatives of patients with OCD also suffer, since symptoms often result in restricted access to certain rooms, involvement of others in compulsive rituals, and difficulty in enjoying life (e.g., the inability to enjoy a family vacation because of the sufferer's

obsessions and rituals; Black, Gaffney, Schlosser, & Gabel, 1998). A diagnosis of OCD is also related to increased (non-mental-health) medical utilization.

Clinical Implications of Comorbid Conditions or Mitigating Factors

Accordingly, at the beginning and end of treatment, therapists should assess for associated features of OCD such as mood and other anxiety disorders, interpersonal functioning, academic and occupational performance, general quality of life, and physiological indices such as sleep and sexual functioning. Assessment of these variables before beginning treatment helps the therapist plan for therapy that addresses, as comprehensively as possible, the clinical problems frequently associated with OCD. The same assessment at the end of treatment might also help therapists prepare patients who might be at greater risk for relapse, especially since some of the associated features, such as poor family functioning and mood disturbance, are associated with greater OCD symptom severity.

EPIDEMIOLOGY

Once considered a rare condition, OCD is known to affect from 2–3% of the adult population across the world (Ruscio, Stein, Chiu, & Kessler, 2010). The disorder is slightly more common among females than males. OCD onset is in childhood or adolescence, typically begins by the age of 25, and only rarely after age 50 (Rasmussen & Tsuang, 1986). The mean age of onset tends to be earlier in males (about 21 years of age) than in females (22–24 years; Lensi et al., 1996).

Although most individuals with OCD do not identify clear-cut precipitants to symptom onset, stressful or traumatic events and experiences may play a role for some patients (Gershuny, Baer, Radomsky, Wilson, & Jenike, 2003). Symptoms also occur at higher than expected rates among childbearing women (Abramowitz, Schwartz, Moore, & Luenzmann, 2003; Russell, Fawcett, & Mazmanian, 2013). Moreover, the content of obsessional thoughts and rituals is clearly influenced by environmental and sociological factors, as evidenced by the predominance of obsessions concerning timely issues. For example, the highly prevalent contamination obsessions relating to gonorrhea in the 1970s have largely been replaced by fears of acquiring AIDS since the mid-1980s. Similarly, individuals with religious obsessions show concerns related to their specific religion (e.g., Jewish patients afraid of violating the dietary laws associated with that religion).

COMORBIDITY

Individuals with OCD are at an increased risk for additional forms of psychopathology, with depressive disorders among the most commonly co-occurring difficulties. The lifetime prevalence of major depressive disorder (MDD) among patients with OCD ranges from 12.4 to 60.3% cross-nationally (mean = 29%; Weissman et al., 1994) and, for the most part, OCD predates MDD, suggesting that depressive symptoms usually occur in response to the distress and impairment associated with OCD.

Depressive symptoms also seem to be more strongly related to the severity of obsessions than to severity of compulsions (Ricciardi & McNally, 1995). Also, coexisting depressive symptoms are especially predictive of poor quality of life for those suffering with OCD.

OCD may also be accompanied by additional anxiety disorders (e.g., Weissman et al., 1994), with social phobia, panic disorder, and GAD being most commonly reported (Nestadt et al., 2001). Estimates of comorbidity between OCD and personality disorders vary widely (from 8.7 to 87.5%), yet those personality disorders belonging to the anxious cluster (e.g., obsessive–compulsive, avoidant) are more common than those of other clusters (e.g., Steketee, Eisen, Dyck, Warshaw, & Rasmussen, 1999). We present the clinical implications of comorbid conditions, especially the impact of mood and personality disorders, later in the chapter.

ETIOLOGY

Biological Models

The Serotonin Hypothesis

The serotonin hypothesis was the leading neurochemical theory of OCD. It proposed that obsessions and compulsions arise from abnormalities in the serotonin neurotransmitter system, specifically, a hypersensitivity of the postsynaptic serotonergic receptors (Zohar & Insel, 1987). The most consistent findings in support of this model came from the pharmacotherapy literature, which suggests that selective serotonin reuptake inhibitors (SSRIs; e.g., fluoxetine) are more effective than medications with other mechanisms of action (e.g., norepineherine inhibitors) in reducing OCD symptoms. Actual studies of biological markers—such as blood and cerebrospinal fluid levels of serotonin metabolites—however, largely do not support a relationship between serotonin and OCD (Bandelow et al., 2016). As a result of these findings, contemporary biological theorists have all but abandoned the serotonin hypothesis of OCD.

Structural Models

Structural models hypothesize that OCD symptoms are caused by neuroanatomical abnormalities in localized brain regions, specifically the orbitofrontal–subcortical circuits (Saxena, Bota, & Brody, 2001), which are thought to connect regions of the brain responsible for the initiation of behaviors implemented with little conscious awareness. There is, however, a lack of consistent evidence to support this hypothesis (e.g., Bandelow et al., 2016). These models were derived from neuroimaging studies comparing brain function between people with and without OCD (for a review, see Whiteside, Port, & Abramowitz, 2004). Although findings vary across studies, a meta-analysis of 10 positron emission tomographic (PET) and single-photon emission computed tomographic (SPECT) studies found that relative to healthy individuals, those with OCD evidence more activity in the orbital gyrus and the head of the caudate nucleus (Whiteside et al., 2004).

Limitations of Biological Models

There are important limitations of biological models of OCD, which render these hypotheses untenable. First, there is little correspondence between the patterns of symptoms that patients report and the biological mechanisms proposed to account for them; that is, there is no sufficient explanation for how neurotransmitter or neuroanatomical abnormalities translate into obsessions and compulsions but not into other psychological symptoms such as eating disorder symptoms (e.g., body image disturbance). Second, biological models are unable to explain the fact that OCD symptoms concern some themes (e.g., fears of harming vulnerable individuals), but not others (e.g., fears of harming those who could defend themselves). Also, why do some people experience contamination-related obsessions, whereas others have intrusive sexual obsessions? Finally, existing serotonin and neuroimaging studies are correlational in design and therefore do not provide evidence that biological abnormalities *cause* OCD symptoms.

Psychological Models

Learning Theory

Behavioral (learning and conditioning) models of OCD were based on Mowrer's (1960) two-stage theory of fear acquisition and maintenance. Obsessional fears were thought to be acquired (Stage 1: classical conditioning) as a result of traumatic conditioning experiences in which a previously neutral stimulus (the conditioned stimulus, or CS) was paired with an aversive stimulus (the unconditioned stimulus, or UCS; e.g., a traumatic experience), so that the CS comes to elicit a conditioned fear response (CR). As a result of this associative learning process, specific situations (e.g., using a public bathroom), objects (e.g., door knobs), and thoughts, images, doubts, or impulses (e.g., incestuous images) that pose no objective threat may come to evoke obsessional fear. Stage 2 of the model (operant conditioning) proposes that avoidance behaviors and compulsive rituals become habitual because they reduce obsessional anxiety immediately (i.e., negative reinforcement). By the 1980s, however, limitations of learning models had become apparent. Specifically, many people with OCD did not describe the types of classical conditioning experiences that were thought to lead to obsessions. Thus, additional models were sought to explain obsessional thinking.

Cognitive-Behavioral Models

Most theorists and researchers now apply Beck's (1976) cognitive specificity theory of emotion to understanding the development and persistence of OCD. This model proposes that emotional distress—such as anxiety and fear—arises not from negative or threatening situations and stimuli themselves, but from how one *interprets* or *appraises* such stimuli. Accordingly, obsessional fears are thought to arise from inaccurate beliefs about the importance of intrusive thoughts. Indeed, unpleasant mental intrusions (i.e., thoughts, images, and impulses that intrude into consciousness) are a normal and universal experience for people with and without OCD (Rachman & de Silva, 1978). Rachman (1997) and Salkovskis (1996) have proposed that these mental

intrusions develop into clinical obsessions when they are *misappraised* as overly significant, threatening, harmful, or as posing a threat for which the individual is personally responsible.

A number of similar cognitive-behavioral theories have been proposed, each emphasizing a slightly different type of belief or appraisal of intrusive thoughts. Salkovskis (1996), for example, emphasized inflated responsibility as a key cognitive bias. Rachman (1997) emphasized the overimportance of thoughts. Clark (2004) emphasized the need to control intrusive thoughts as the main dysfunctional belief. A comprehensive model that incorporates all of these beliefs and appraisals was developed by the Obsessive Compulsive Cognitions Working Group (e.g., Frost & Steketee, 2002), which identified six domains of dysfunctional beliefs that play a role in OCD, as described in Table 5.3. To illustrate, consider an unwanted image of murdering a loved one. Most people would regard such a thought as "mental noise" (i.e., having no harm-related implications). Such an intrusion, however, might develop into a clinical obsession if the person appraises it as having serious consequences, for example, "This thought means I am a dangerous person." Such appraisals evoke anxiety and the urge to suppress or remove the unwanted intrusion and the anxiety using compulsive rituals or other neutralizing strategies, and to avoid reminders of the thought (e.g., by locking away all knives), to minimize worries about disasters.

Salkovskis (1996) proposed that compulsive rituals and avoidance develop into strong patterns because they seem to work immediately (albeit temporarily) to reduce distress and remove the unwanted obsessional thought, and are therefore negatively reinforced. Such behaviors, however, also prevent the person from discovering that the feared outcome (e.g., acting on a senseless intrusive thought) is highly unlikely

TABLE 5.4. Domains of Dysfunctional Beliefs Associated with OCD as Identified by the Obsessive–Compulsive Cognitions Working Group

Belief domain	Description
Excessive responsibility	Belief that one has the special power to cause and/or the duty to prevent negative outcomes.
Overimportance of thoughts	Belief that the mere presence of a thought indicates that the thought is significant. For example, the belief that the thought has ethical or moral ramifications, or that thinking the thought increases the probability of the corresponding behavior or event.
Need to control thoughts	Belief that complete control over one's thoughts is both necessary and possible.
Overestimation of threat	Belief that negative events are especially likely and would be especially awful.
Perfectionism	Belief that mistakes and imperfection are intolerable.
Intolerance for uncertainty	Belief that it is necessary and possible to be completely certain that negative outcomes will not occur.

in the first place. Thus, the mistaken interpretation of the thought persists, and the obsessional fear is maintained.

Treatment approaches derived from the biological and psychological models are reviewed in the following section on treatment outcome.

TREATMENT OUTCOME

Cognitive-behavioral models hold the greatest promise for understanding and treating OCD symptoms, as they are the only approaches that are empirically supported and provide a logically consistent basis for treatment. In particular, from this perspective, successful treatment for OCD must (1) correct maladaptive beliefs and appraisals that lead normal intrusive thoughts to escalate into obsessional fear and (2) terminate avoidance and compulsive rituals that prevent the correction of these maladaptive beliefs. This section provides a concise description of two interventions—exposure and response prevention (ERP) and acceptance and commitment therapy (ACT)—that are consistent with cognitive-behavioral models and can be used on their own or in tandem to reduce OCD symptoms. Detailed guidelines for their use are available in various treatment manuals (e.g., Abramowitz & Jacoby, 2014).

Exposure and Response Prevention

Exposure therapy comprises a set of psychological treatment procedures designed to help individuals confront stimuli that provoke obsessional fear, but that objectively pose a low risk of harm. Exposure for OCD can occur in the form of repeated encounters with the actual feared situations (situational exposure) and imaginal confrontation with the feared disastrous consequences of confronting these situations (imaginal exposure). For example, an individual with obsessional thoughts of harming children might for situational exposure be asked to go to playgrounds and schools where there are small children. The individual might also practice imaginal exposure to images of harming children. A patient with fears of contamination from disabled persons might for situational exposure shake hands with a visually impaired or hearing-impaired individual. For imaginal exposure, the patient would then confront thoughts of losing their vision or hearing as a result of the encounter. During exposure tasks, individuals are helped to put their fear predictions to the test in a range of contexts and using varied fear stimuli, so they can learn that their obsessional triggers are in fact safe. They are also helped to learn that obsessional anxiety and uncertainty are safe and manageable, often by observing the naturally occurring process of habituation in which anxiety temporarily increases followed by a natural decline in distress.

The response prevention component of treatment entails refraining from compulsive rituals and other strategies that serve as an escape from obsessional fear. Response prevention helps to prolong exposure and facilitate safety learning. In the earlier examples, the first patient might practice refraining from any strategies they typically use to reassure themself that they won't act on thoughts of harming children. The second patient would be instructed to refrain from decontamination rituals such as washing.

Most ERP treatment manuals (e.g., Abramowitz & Jacoby 2014) recommend that ERP be conducted over the course of 12–20 sessions. The initial few sessions include a thorough assessment of obsessions, compulsive rituals, avoidance strategies, and anticipated consequences of confronting feared situations. This information is used to plan the specific ERP exercises that will be pursued. The therapist also never physically restrains the individual from performing rituals; rather, the therapist helps the person to resist these behaviors. Exposure exercises may progress gradually using a hierarchy—a list of feared situations and thoughts arranged from the least to the most anxiety provoking—although this not required. Beginning exposure with less anxiety-evoking tasks increases the likelihood that the patient will learn to manage distress and fosters confidence in the treatment. At the end of each treatment session, the therapist instructs the patient to continue exposure for several hours and in different environmental contexts, without the therapist.

Numerous studies involving thousands of individuals with OCD affirm that ERP is an effective treatment for this condition. Randomized controlled trials (RCTs) show the superiority of ERP over placebo treatments such as progressive muscle relaxation training (e.g., Fals-Stewart, Marks, & Schafer, 1993), anxiety management training (Lindsay, Crino, & Andrews, 1997), and pill placebo (Foa et al., 2005). RCTs also show that ERP is more effective than the antidepressant clomipramine, believed to be the most effective form of pharmacotherapy for OCD (Foa et al., 2005). Although patients receiving ERP typically show a 50–60% reduction in their OCD symptoms, some residual symptoms usually remain even posttreatment (Olatunji, Davis, Powers, & Smits, 2013).

Acceptance and Commitment Therapy

ACT (Hayes, Stroshal, & Wilson, 2011) is a set of techniques that can be applied in the treatment of OCD to help individuals foster a willingness to experience their obsessional thoughts, uncertainty, and anxiety, and respond flexibly in the presence of these unwanted private events (Twohig et al., 2015). ACT is also used to foster improvements in quality of life and the pursuit of things the patient values in life. The ACT perspective on OCD is consistent with the cognitive-behavioral model previously presented, as well as with ERP—in fact, one can implement ACT along with ERP (e.g., Twohig et al., 2015). While an important goal of both ACT and ERP is to broaden the patient's engagement with feared stimuli and improve quality of life, ACT explicitly focuses on (1) fostering willingness to experience obsessional distress, (2) recognizing thoughts and feelings as neither right nor wrong (i.e., "cognitive defusion"), and (3) using treatment to move toward what one values in life. Most ACT-based techniques take the form of metaphors that are discussed in the context of OCD and its treatment, and are referred to throughout treatment to revisit the processes described next.

Fostering Willingness

Willingness, in this context, refers to being open to "experiencing your own experiences" without trying to change, avoid, or escape them. Metaphors, such as "two scales," "jerk at the door," and "tug of war with a monster" (e.g., Abramowitz & Jacoby,

2014) can be used to help the patient understand the goal of treatment as developing a healthier (more peaceful) relationship to OCD-related inner experiences.

Defusing from Thoughts and Feelings

Defusion strategies help people with OCD give less power to their anxiety-provoking thoughts, feelings, and body sensations. More specifically, patients are helped to shift away from evaluating these and other internal experiences as objectively "good," "bad," "dangerous," "immoral," and so forth. Put another way, the aim is to use metaphors (e.g., "passengers on the bus," "becoming the chessboard"; e.g., Abramowitz & Jacoby, 2014) to help the individual see obsessions and anxiety for what they really are (streams of words, passing sensations), not what they say they are (facts or dangers). Patients get to decide for themselves how much weight to give to particular experiences.

Moving toward Values in Life

Improving patients' quality of life is an important outcome of ACT, and a number of exercises can be used to help patients realize that the hard work of treatment is worthwhile.

Research on the efficacy of ACT for OCD is in its nascent stages, but preliminary evidence is encouraging. Twohig and colleagues (2010) found that ACT is superior to muscle relaxation at posttreatment and at follow-up, with between-group effect sizes of 0.77 and 0.62, respectively. Nevertheless, when Twohig and colleagues (2018) compared ACT-based ERP to standard ERP, there was no evidence that ACT enhances the effects of ERP. Thus, further research on the effects of ACT and how it may best be used in the treatment of OCD is warranted.

Hypothesized Mechanisms of Change in Psychological Treatments

From a learning perspective, ERP provides an opportunity for the extinction of conditioned fear responses. Traditionally the efficacy of ERP was understood in terms of emotional processing theory (EPT; Foa, Huppert, & Cahill, 2006; Foa & Kozak, 1986), which credits initial fear activation followed by habituation as the mechanisms of improvement. The basic assumptions of EPT, however, are not well supported by the research evidence (Craske, Treanor, Conway, Zbozinek, & Vervliet, 2014): Habituation during ERP often fails to predict long-term outcomes, and successful outcomes may occur in the absence of habituation.

Numerous developments in research on learning and memory relevant to exposure and extinction point to *inhibitory learning* as the mechanism of extinction (e.g., Craske et al., 2014). From the inhibitory learning perspective, the original danger-based association between the conditioned and unconditioned stimulus (e.g., "Floors cause sickness," and "Uncertainty about the future is intolerable") remain intact during ERP, while competing non-danger-based associations (e.g., "Floors are generally safe," "Uncertainty is manageable") are formed.

Increasing willingness to experience fear and uncertainty, as in ACT, complements the goal of inhibitory learning. To the degree that distress is accepted (e.g.,

"Anxiety is a part of life, so I'm better off learning to accept it rather than trying to fight it"), inhibitory associations can be more robustly acquired ("If fear is tolerable, then I can push myself to try exposure therapy and learn that I can manage this acceptable risk"). Increasing tolerance for experiences such as fear, disgust, and uncertainty also reduces the likelihood that an inevitable unexpected encounter with one of these experiences in a new context (e.g., once treatment has ended) will lead to the return of fear and a relapse.

Pharmacotherapy

Selective serotonin reuptake inhibitor (SSRI) medication is the most widely used pharmacological treatment for OCD. The specific agents in this class of drugs include fluoxetine, paroxetine, sertraline, citalopram, and fluvoxamine. Clomipramine, a tricyclic medication that possesses serotonergic properties, is also used in the treatment of OCD. On average, relative to placebo, SSRIs produce a 20–40% reduction in obsessions and compulsions (Rauch & Jenike, 1998). The major strengths of a pharmacological approach to treating OCD include the convenience and the little effort required on the individual's part. Limitations include the relatively modest improvement and likelihood of residual symptoms, high rate of nonresponse (40–60% of patients do not show any favorable response), and the prospect of unpleasant side effects (which can often be stabilized by adjusting the dose). Moreover, once SSRIs are terminated, OCD symptoms typically return rapidly (Pato, Zohar-Kadouch, Zohar, & Murphy, 1988).

CLINICAL IMPLICATIONS OF BASIC RESEARCH FOR TREATMENT

Various aspects of the psychopathology of OCD have direct treatment implications and can be used to establish informal guidelines regarding the use of ERP and medication, or their combination. We discuss these implications in this section.

Poor Insight

Research consistently demonstrates that relative to individuals who recognize their obsessions and compulsions as illogical and senseless, those with poor insight into senselessness of their symptoms show attenuated response to both ERP and pharmacotherapy (e.g., Foa et al., 1999). It appears that patients with poor insight have difficulty consolidating information that is inconsistent with their obsessional fears; thus, their beliefs and expectations of danger are not modified with repeated exposure to obsessional cues. Moreover, because of their extreme fear, these patients might not engage in ERP tasks as well as do those with good insight. As a result, they do not gain opportunities to correct strongly held inaccurate beliefs and appraisals. The reason that poor insight hinders response to medication is not well understood, yet the fact that both of these treatments are attenuated by poor insight suggests that augmentation strategies are necessary. In the case of psychological treatment, some authors suggest the use of cognitive therapy approaches to help with verbally changing beliefs and appraisals rather than using exposure alone (McKay et al., 2015).

Severe Depression

Between 25 and 50% of patients with OCD suffer from comorbid depressive disorders (Abramowitz, 2004), and studies that have examined the effects of depressive symptoms on the outcome of ERP have revealed that relative to mildly and moderately depressed patients with OCD, those with severe depression—and indeed a diagnosis of MDD—have attenuated outcomes with ERP. Perhaps because of their high emotional activity, severely depressed individuals fail to undergo a reduction in anxiety/distress (i.e., habituation) that normally occurs following extended exposure to feared stimuli. Thus, depressed patients with OCD may not learn how to feel comfortable in the presence of obsessionally feared situations. Motivational difficulties, which often accompany depression, may also account for poor treatment outcome; that is, patients may not adhere to instructions to face their fears and resist urges to ritualize if they are extremely depressed and unmotivated to help themselves. Some research points to the use of antidepressant medication to help depressed patients benefit from ERP (e.g., Simpson & Liebowitz, 2005). Other authors have proposed that the use of cognitive therapy for both OCD and depression may enhance outcome for this group of patients (Abramowitz, 2004).

Personality Disorders

The presence of certain personality traits and personality disorders is known to hinder response to ERP and medication for OCD (Steketee, Chambless, & Tran, 2001). OCPD and histrionic personality disorder traits, for example, might interfere with developing rapport and adherence to ERP instructions. If, however, a good therapeutic relationship can be developed, ERP can be successful despite these traits. Clinicians should also consider that some patients with dramatic traits gain reinforcement for their OCD symptoms. In such circumstances, psychological treatment approaches are unlikely to succeed because patients do not perceive themselves as gaining rewards for their efforts to reduce obsessions and rituals. Individuals with personality traits in the odd cluster (e.g., schizotypal personality disorder) present a challenge for clinicians because such patients have a reduced ability to profit from corrective information obtained during exposure or cognitive interventions.

Clinical Implications for Treatment Planning with Different OCD Symptom Presentations

As described earlier in this chapter, the symptoms of OCD vary considerably. Below are some specific treatment implications for particular presentations of this clinical disorder.

Contamination Concerns

Contamination obsessions and decontamination rituals respond well to ERP involving planned exposure to sources of contamination, contaminated objects, or "dirty" environments. Response prevention can pose a challenge, especially if the patient fears situations such as using the bathroom, which will be confronted multiple times

per day. Ideally, response prevention should include no washing at all, although exceptions for bathroom use and before meals can be made initially (until such items are confronted for exposure). If the patient is reluctant to end all washing, a limited amount of ritualizing may be permitted per day, with gradual reductions over time. It is important for the therapist to attend closely to the actual washing methods, which may be lengthy, detailed, or completed according to a set of "rules."

Order and Symmetry Concerns

This symptom dimension often responds well to ERP, which emphasizes the response prevention component, because rather than specific obsessions, there may be only anxiety about "not feeling right" unless rituals are carried out. Exposure may involve deliberately leaving things "not just right"—pictures hung crookedly, books arranged in a haphazard manner, doors left open, and so forth—in which case the urge to complete the associated ordering or arranging rituals is resisted. The focus is on tolerating the anxiety and "not just right" feelings, and finding out they diminish with time, even without performing compulsive rituals.

Responsibility Obsessions and Checking Rituals

Responsibility obsessions often require the use of both situational exposure to feared situations and imaginal exposure to feared consequences. For example, a woman afraid of hitting pedestrians while she is driving, and who checks the roadside for bodies, might perform situational exposure to driving in areas with high pedestrian traffic (e.g., parking lots) and subsequently imagine that she hit someone but did not realize it. Instead of acting on her urges to return to the parking lot to check, she would be helped to use response prevention to remain exposed to her uncertainty and learn that her obsessional anxiety eventually subsides. ACT might also be used directly to address the tendency to resist feelings of guilt and respond as if inflated perceptions of responsibility are accurate.

Mental Rituals

It has long been thought that mental rituals, which are often accompanied by disturbing obsessional thoughts about violence (e.g., images of stabbing loved ones), sex (e.g., doubts about whether one is becoming gay), and religion (e.g., blasphemous ideas), are the most difficult presentation of OCD to manage with psychological treatments. In addition to the typical anxiety and fear of acting on intrusive thoughts that seem senseless, individuals with these types of symptoms often become concerned that their obsessions reflect their true nature (e.g., "Why would I be thinking about this if I didn't really feel this way?"). It can be helpful for patients to know that "bad thought OCD" tricks them in three ways: It says this thought is important, it means something about them as a person, and the thought requires a response. Also, their more typical thoughts, even if about difficult content, are not accompanied by a wave of anxiety or distress.

An important goal for the clinician is to assess carefully rituals and other strategies used to respond to such obsessions, and to avoid falling into the trap of labeling

such individuals as "pure obsessional" even if no rituals can be observed. Most likely, patients are using covert (mental) rituals and brief neutralizing strategies in response to their obsessions. Mental rituals can take many forms, such as replacing a "bad" thought with a "good" one, mentally blocking a "bad" thought by imagining an *X* through it, saying a prayer, repeating magic words or phrases to oneself ("God keep them safe"), or mentally reviewing one's behavior for evidence of the trait concerned about. One of the most problematic rituals or reactions for patients is when they check for or believe they experience an arousal reaction in their groin to sexual intrusions.

Exposure therapy involves confronting stimuli that trigger obsessional thoughts (e.g., places of worship, knives), as well as imaginal confrontation with the obsession itself, and the feared consequences of not ritualizing (e.g., thoughts and images of molesting children). Imaginal exposure tasks usually involve first writing scripts of the patient's obsessional fears that incorporate varying amounts of detail to modulate anxiety (e.g., "I have a thought about my grandmother dying, but I don't say a prayer for her. As a result, she dies, it is all my fault, and my family abandons me"). The patient is then helped to record and listen to the scenario, or simply to read it repeatedly until habituation occurs (i.e., the scene becomes "boring" or does not provoke anxiety). In addition, the patient is helped to refrain from rituals that would artificially reduce the obsession and the resulting anxiety. ACT techniques can be used to further help the patient defuse from obsessional thoughts and learn how to persist with valued activities without allowing these obsessions to get in the way.

Engaging Patients, Motivation, and Collaboration

Most psychological therapies require engaging patients and motivating them for treatment, but this is especially true in the treatment of OCD when trying to convince patients to confront anxiety-provoking thoughts, take perceived risks, and not engage in behaviors that they believe are essential for their own (or others') safety. One thing that would surprise most clinicians who do not work in a cognitive-behavioral format is how much education alone can help with engagement and motivation. Patients often say to clinicians familiar with OCD, "You really get this" or "You know what I am talking about," just by the questions asked and the responses provided by the knowledgeable therapist. This knowledge increases the therapist's credibility and increases hope that recovery is possible.

Especially with OCD symptoms that can seem "crazy," it is comforting that there is some explanation about how patients get stuck and why. Fortunately, education about how OCD works both teaches the patient about the problem and leads directly to how to treat it. Education is essential before any treatment can begin. Talking with patients about the fact that they have a low tolerance for uncertainty and respond more negatively to that uncertainty or intrusive thoughts often resonates with them. Also discussing how rituals begin and are maintained—because they work in the short run by decreasing anxiety—teaches patients why they feel better right after they do their rituals. Explaining that everyone would do rituals if they felt it was the only way to keep their family safe or protected expresses empathy and understanding of what the client is experiencing. Indeed, rituals may seem like a small price to pay if they were the only way and they worked. However, patients themselves know and report that relief is short lived, and the concern repeats and repeats.

The cruel trick of OCD is that performing the ritual makes obsessional thoughts occur more frequent and the ritual feel more necessary. The short-term relief is traded for longer-term loss of freedom and interference with living. The other lie of OCD is that the intense distress the patient experiences after the obsession will continue on—that day, that night, or longer, if the ritual is not performed. Cognitive-behavioral therapists talk to patients about retraining their brains to respond differently to the error message or danger warning, and to break the cycle between the ritual and relief. While education is not enough, therapists who use it are continually reminded of how important it can be. This might be especially true of OCD because so little about it seems to make sense, and patients not only know they are doing "crazy" things but also know how difficult it feels to stop.

One important aspect of motivation is discussing the loss of freedom and time, and ascertaining what the patient is missing because of OCD. They may lose time for favored activities, sleep, enjoying a meal, being able to touch or hug a family member, travel, and so forth. Each person is different, but it is important to have incentive to start this effective but challenging treatment, and it has to be the patient's own motivation. Motivational interviewing, formal or informal, can be helpful to discuss the reasons whether to change or not. It is important to be clear that it is fine to be ambivalent and afraid, and most clients start that way. It is also crucial to talk to patients about what they believe about the obsessions when in a state of calm, versus in an OCD crisis moment.

The therapist sets the tone of a collaborative venture in discussing, planning, and designing treatment goals. Patients are told that they are in charge of the pace and content of their exposures, since they cannot and will not be forced to do anything. However, the therapist will be using their knowledge to guide treatment, to recommend and select goals, and to push patients past their comfort zone. Indeed, patients need to understand that if they do not feel anxiety or distress at the beginning of treatment and exposures, the treatment is not working. The goal is to start with manageable levels of anxiety and easier tasks, and build competence and confidence, session by session. It can be helpful to tell patients that most people with OCD do not feel it until they see it for themselves by doing the treatment. Patients need to experience that the distress goes away, lowers or at the least is tolerable.

While education is very important, clinical experience suggests that patients only begin to believe in the treatment of OCD and hope for recovery when they do something different relative to their OCD. What patients with OCD need, in other words, are corrective experiences, particularly about the nature of anxiety and their ability to tolerate it. In contrast, patients with OCD can be experts at engaging therapists in endless discussions and questions about their fears and obsessions. The anticipation and concern about exposure is often the most difficult part, so it is important to pick a small threat exposure (something as small as writing K_ _ _ for kill) soon and get to it rather than have anxiety build and allowing avoidance by more explaining.

A more supportive, exploratory "talk-oriented" therapy can get hijacked by OCD, with therapists' spending session after session providing reassurance, answering questions, reviewing past events, or trying to be logical about OCD. While the treatment of other anxiety-related problems may benefit from supportive therapy or logical analysis of thought styles or patterns, OCD rarely does. Indeed, what is

helpful is to perform early exposures with therapist supervision so that the patient can understand and experience what exposure is like, see how to address issues that arise, and modify tasks or goals as needed.

All of these components of treatment—education, defining the motivation, and collaboration—are essential for all therapeutic approaches to be successful. These components are a basis for a good therapeutic alliance. These factors, in addition to empathy and soliciting regular feedback, about the working relationship have been found to be associated with good outcomes for empirically supported treatments across multiple disorders. These relationship qualities are likely to be especially essential when treatment of OCD requires clients to take risks in facing fears, and to trust the process and the therapist to do so. Many therapists who treat OCD have had clients say, "I did not like this process or how you pushed me but it worked and I am glad you did." Compliance, an additional treatment variable with in-session and out-of-session tasks, is crucial to the success of OCD treatment and likely is far more important than in the treatment of other anxiety disorders or other disorders overall.

Exposure and Response Prevention

The only way to do ERP is to face feared situations and to eliminate rituals. In other words, a cognitive-behavioral therapist might distinguish between providing therapy and treating OCD. Behavioral treatment of OCD cannot involve just talking; it has to include doing. Rather than being asked, "What can't you do?" the patient is often asked, "What can you do?" Doing nothing is not an option for recovery. However, any small step can provide an opening.

For all OCD thoughts, but especially for intrusive bad thoughts, OCD therapists are teaching clients to respond differently to the presence of a thought. Simply put, we want to have patients label thoughts as just thoughts and nothing more. OCD often presses patients to feel they have to respond to them, do something, or get rid of them. So we teach patients to let OCD thoughts go in and go out, as they do with other thoughts. Do not make efforts not to think or block thoughts. That does not work, or it makes them come more frequently. But, also, do not hang on to the thought through actions physical or mental (use it as reason to check, analyze evidence for the bad thought, review the day, or replace it with a good thought). The ACT approaches described below can also help with this for ERP, or as an alternative treatment approach.

While the goal of the treatment of OCD is to eliminate rituals, there are a number of ways to address this other than full-scale elimination if this is not possible. This is where the protocol-limited research programs and independent clinical work may diverge. Often, after the education component is completed, the patient who is not ready to begin exposure may be "deemed inappropriate for the treatment program." Some research-based programs require full adherence to a planned limitation of rituals (e.g., a week of no washing or limited showering); however, only some patients can do this, and other options are available for the private practice clinician. Actually, the patients who agree readily to treatment and jump in are often the easiest to treat. It is important to tell patients that while it may be more difficult and anxiety provoking to try to eliminate rituals (e.g., washing only before meals or after

going to the bathroom), they will get faster relief. While the former may be preferable, a more gradual approach is workable, and more patients may be able to comply.

For all OCD symptom presentations, there are ways to delay or modify rituals. Many patients are more willing to consider these as first steps to delay a ritual (e.g., delay checking for 20 minutes). The other option is to pick one period of day to be OCD free (e.g., "From 9:00 to 12:00 P.M., I won't rearrange things in my kitchen"). Also, a patient can work to reduce the frequency of the ritual (limit reassurance questions to 10 rather than usual 30 per day). Finally, a patient can "do OCD wrong" (e.g., use two squirts of soap instead of four, or do their full OCD prayer ritual but in a different order).

An important component of safety learning is having a plan for what to do when anxiety persists during and after ERP sessions. Rather than white-knuckling it through the experience, or trying to fight the anxiety, patients should engage in daily activities. This helps them learn that uncertainty, anxiety, and obsessional thoughts are manageable experiences that do not have to bully the person into avoidance or interrupt or limit their activities. Also, clients need to have their OCD treatment plan in mind or nearby, knowing how to react rather than just waiting for it to arise and being surprised. This often involves written notes, reminders on phones, and so forth, since being able to remember is difficult under stress and especially in the beginning

Implications of Client Variables for Treatment Planning

As in the predictors of outcome for other anxiety disorders, severity of disorder, comorbidity with other psychological disorders, and overvalued ideation of OCD thoughts are predictors of poorer outcome in OCD treatment. It may be that these patients show slower progress, need more sessions, set more modest goals, and demonstrate more modest outcomes. However, clinically, we have found with OCD that severity may interact with motivation, such that if the OCD is very distressing or interfering, patients are motivated to reclaim their lives or "not want to continue like this anymore." If the client is willing, it may be helpful to attempt treatment even with comorbid conditions because OCD is often easier to treat and quicker to show treatment response, which can help with hope and motivation. Very serious depression or substance use may require alternative psychosocial and/or medication treatments before the person has enough energy and focus to engage in the challenging work of ERP. Overvalued ideation may require more cognitive therapy before ERP can begin. Finally, demographic variables have not been strong predictors of response to treatments for anxiety.

Group Therapy

A few studies have examined the effectiveness of group treatment (ERP and cognitive therapy) for OCD (McLean et al., 2001). One strength of a group approach is that patients can get support and understanding, and learn that they do not suffer alone. Education about OCD can be delivered in a didactic format, since the general cognitive-behavioral model of OCD applies to patients across symptom dimensions. However, the symptom heterogeneity in OCD also limits this approach. Specifically,

different patients in the group may require individual ERP plans that cannot practically be implemented in group format, or even within a group therapy room. Still, patients may experience vicarious learning by watching others engage and manage feared situations, and help their cohorts to challenge mistaken beliefs. Patients can also provide support and encouragement to each other. There are also consumer-led support groups that supplement professional treatment or help with relapse prevention.

Partner or Family Involvement in Symptoms

Individuals with OCD who live with a partner (e.g., spouse) or relatives often involve these other people in obsessions, avoidance, and compulsive rituals. For example, a man with contamination obsessions required that his family take showers with a specific type of soap before coming into his bedroom. In a less direct manner, OCD can affect friends and coworkers. People interacting with individuals with OCD often comply with requests to follow rituals, provide reassurance, and engage in avoidance strategies to keep their loved ones from experiencing anxiety. Because such accommodation is akin to OCD sufferers performing rituals by themselves, it must be targeted in treatment, and the extent to which others can be helped to cut down on such behavior is a predictor of more successful treatment outcome. Indeed, family accommodation is one of the strongest predictors of symptom severity and treatment outcome for both child and adult OCD. This is not that family behavior is the cause of the OCD, but it certainly can maintain OCD. Families and spouses can have a very positive impact on recovery. Additional suggestions for treating OCD in the context of family accommodation can be found in Abramowitz and colleagues (2013). In a more global manner, low levels of social support or, in contrast, high levels of hostility and criticism, are negative predictors of outcome and influence relapse across multiple disorders, including anxiety disorders. Therefore, interventions that increase social support and decrease negative social interactions are likely to impact treatment outcome. These interventions need to be "responsive." Some families are very supportive and have low levels of OCD accommodation. Other family work is crucial for treatment success to address impediments that are clearly present and impactful.

Relapse Prevention

For the majority of people with OCD, treatment produces noticeable improvement in symptoms, but residual obsessional fears, avoidance, and compulsive urges usually remain (Abramowitz, 1998). In addition, these residual symptoms are likely to wax and wane over time, often becoming more intense during periods of stress (e.g., relationship difficulties, job loss). Thus, any improvement attained during treatment should be carefully monitored to ensure that temporary lapses do not turn into full-blown relapses. Relapse prevention programs for OCD using ERP and cognitive therapy have been developed and tested (e.g., Hiss, Foa, & Kozak, 1994), and can be used as a way to taper off treatment.

An important part of preventing relapse is recognizing that obsessions and compulsions may be inevitable. Thus, the patient should not be surprised or frightened

by such symptoms when they occur. Cognitive therapy techniques can be used to modify any maladaptive beliefs about the recurrence of symptoms (e.g., "It's my fault that I'm slipping," "I'll be back to how I was before treatment"). Psychoeducation helps patients understand that OCD is usually a chronic problem, that they should expect symptom exacerbation during times of life stress, and that they should interpret symptoms within this framework and use the skills they have learned during treatment to manage these residual problems. Relapse prevention also teaches patients to recognize when new or different obsessions and compulsions arise, and how to view these as similar to symptoms that may already have been addressed by recognizing any rule or "have to" behavior is OCD and to recognize the function of the behavior or compulsion to get rid of anxiety. Indeed, rapid application of ERP techniques can prevent symptoms from escalating or interfering with functioning.

Other Treatment Approaches

While a wealth of research evidence supports the use of ERP, pharmacotherapy, and to a lesser extent, ACT, other forms of psychotherapy have not been evaluated for OCD. Mindfulness approaches, often a part of ACT protocols, provide clients with a way to notice rather than react to their thoughts or feel compelled to action, and may therefore be a helpful tool for addressing OCD. Unfortunately, at times, therapists who do not use empirically supported approaches end up inadvertently reinforcing OCD symptoms. For example, therapists who repeatedly try to provide reassurance to patients with obsessional doubts—instead of helping them face their anxiety and learn to manage obsessional uncertainty and doubt—are helping the patient ritualize by reassurance seeking. Others try to "help" patients overcome obsessions by trying to stop these thoughts (e.g., by snapping a rubber band on their arm when an obsession comes to mind) rather than confront them therapeutically to learn that they are not dangerous. This "thought stopping" technique, however, is akin to mental rituals such as thought suppression, which, as we have discussed, do not work in the long run (i.e., obsessions return, especially when the person tries to dismiss them). Finally, the first inclination of many therapists when they see anxious and fearful individuals may be to try to soothe them, perhaps by teaching them relaxation strategies or biofeedback. Unfortunately, however, this approach is the opposite of exposure therapy, in which the person is helped to go *toward* anxiety in order to learn that it is a temporary feeling, that, although unpleasant, is not dangerous. Approaches such as these are not consistent with the conceptual models of OCD described earlier; therefore, they are not likely to provide patients with long-term relief from obsessions and compulsions.

CONCLUSION

OCD is a severe and functionally impairing condition with a heterogeneous presentation. Even the most experienced clinicians find themselves at once astounded and baffled by the complexity of the psychopathology, the rich patterns of anxiety-provoking thoughts and anxiety-reducing behaviors, and the severity of the impairments that occur in OCD. Although a definitive cause has not been established,

various biological and psychological models have been proposed. Yet, as with many psychological disorders, more is understood about the *symptoms* of OCD than about its etiology. Fortunately, understanding the symptoms—specifically, that obsessional thoughts provoke anxiety and compulsive rituals are performed to reduce anxiety—has led to tremendous breakthroughs in treatment. Exposure, response prevention, and ACT are all effective strategies for reducing symptoms, although each comes with its own barriers to success. Many patients achieve excellent outcomes that were unheard of before the 1960s, but residual symptoms are still the norm. The field has, however, begun to identify factors that may reliably predict poorer response, such as poor insight into the senselessness of obsessional fears, severe depression, and family hostility.

Still, important topics require further study. For example, patient readiness programs would ensure a lower refusal or premature dropout rate. Treatments that teach partners (and other family members) effectively how to assist with a loved one's ERP would also prove beneficial given the high prevalence of relational problems and symptom accommodation in families of patients with OCD. From the clinical perspective, providing successful treatment for OCD can be one of the greatest challenges a therapist may experience. Yet the fact that most patients show at least some response to contemporary cognitive and behavioral techniques and return to meaningful and value driven lives makes working with this patient population highly rewarding.

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